GEOTECHNICAL ENGINEERING

UNIT - I : INTRODUCTION & INDEX PROPERTIES OF SOILS

Course Code: 23A01503T

Syllabus: Soil formation - Soil structure - Adsorbed water - Mass-volume relationship - Relative density. Index properties: moisture content, specific gravity, in-situ density, grain size analysis (sieve and hydrometer methods), consistency limits and indices - I.S. classification of soils.

1. Introduction

Geotechnical engineering is the branch of civil engineering that deals with the behaviour of earth materials (soil and rock) and their use as foundation support or construction material.

- Soil = unconsolidated, loose or weakly bonded accumulation of mineral particles formed by weathering of rocks, with voids containing air and/or water.
- Rock = hard, naturally formed, firmly bonded mass of mineral matter.
- Soil mechanics = application of the laws of mechanics and hydraulics to soils (Terzaghi - "Father of Soil Mechanics", 1925, Erdbaumechanik).
- Soil is a three-phase system: solids + water + air.

Why study soil?

- Soil is the foundation for every structure: buildings, bridges, dams, roads, runways.
- Soil is the construction material: embankments, earth dams, road subgrade.
- Soil is retained / supported: retaining walls, slopes, excavations, tunnels.

Main areas of Geotechnical Engineering

- Site investigation and soil exploration
- Foundation engineering (shallow and deep foundations)
- Earth pressure and retaining structures, slope stability
- Earth dams, embankments, pavements (subgrade design)
- Ground improvement (compaction, grouting, soil stabilisation, geosynthetics)

2. Soil Formation

Soils are formed by the weathering (disintegration and decomposition) of parent rock by the action of natural agents like temperature, water, wind, ice and living organisms.

(a) Physical (mechanical) weathering - disintegration

- Changes in temperature: day-night expansion and contraction cause cracks and spalling.
- Frost action: water in cracks freezes, expands by about 9% and wedges the rock apart.
- Abrasion by wind, running water, glaciers and wave action.
- Plant roots and burrowing animals widen the cracks.
  - Result: coarse-grained soils - gravel, sand, silt (bulk minerals unchanged).

(b) Chemical weathering - decomposition

- Hydration, hydrolysis, oxidation, carbonation, solution and leaching change the minerals chemically.
- Feldspar + water + CO_2 gives clay minerals (kaolinite, illite, montmorillonite).
  - Result: fine-grained soils - clays (very small particles, large surface area).

(c) Biological weathering

- Lichens, mosses, bacteria, roots and burrowing animals produce acids and organic matter (humus).

Physical weathering produces silt, sand and gravel, while chemical weathering produces clay minerals. Rate of chemical weathering is high in hot and humid climates.

Classification of soils based on origin

Soil type	Transported by	Features / examples
Residual soil	not transported (formed in place)	Grading from fine at top to rock at depth, e.g. laterite, black cotton soil
Alluvial soil	running water (rivers)	Stratified, sand-silt-clay in layers, fertile; Indo-Gangetic plain
Lacustrine	lake deposits	Fine silt and clay, soft, compressible
Marine soil	sea water	Soft marine clays, high salt content, sensitive
Aeolian soil	wind	Loess, dune sand: uniform, fine, loosely packed
Glacial soil	glaciers / ice	Boulder clay (till), very mixed sizes
Colluvial soil	gravity	Talus / landslide debris at foot of slopes

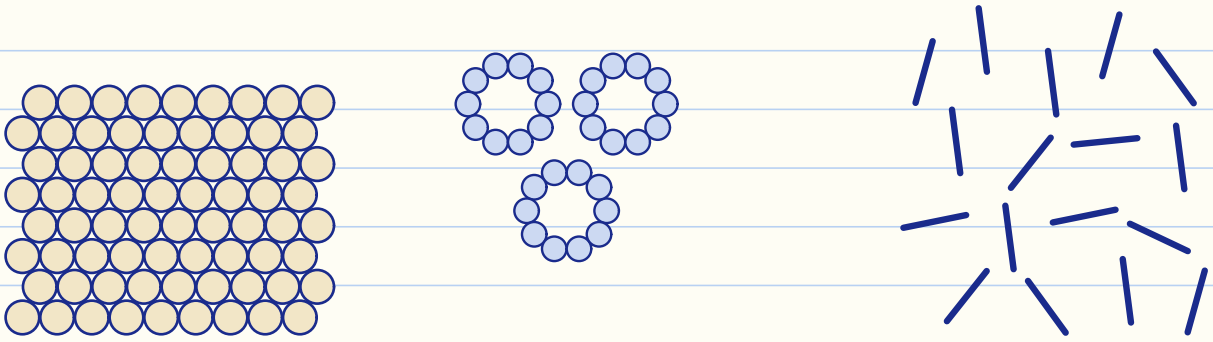
Some Indian soils

- Black cotton soil: montmorillonite, high swelling and shrinkage, Deccan plateau, poor for foundations.

- Laterite soil: rich in iron and aluminium oxides, red, used as road material.
- Alluvial soil: Indo-Gangetic plains and river deltas.
- Desert (aeolian) soil: Rajasthan, very fine uniform sand.

3. Soil Structure

The structure of a soil is the geometrical arrangement of its particles and the forces between them. It governs strength, permeability and compressibility.



(a) Single grained (b) Honeycomb (c) Flocculent

Types of structure

- Single-grained: coarse particles (sand, gravel) settle under gravity; each grain touches its neighbours. Dense or loose packing. Particle weight >> surface forces.
- Honeycomb: fine sands / silts (0.02 - 0.002 mm) settling in water bridge over to form arches around voids. Open structure, high void ratio, can carry static load but collapses under shock.
- Flocculent: clay particles (< 0.002 mm) in salt water. Edge (+ve) of one plate attracts face (-ve) of another giving a "card-house" structure - very high void ratio and strength.
- Dispersed: clay particles in fresh water repel each other, lie parallel to one another - low permeability, low strength after remoulding.
- Coarse-grained skeleton: large grains in contact with fines filling the voids between them.
- Cohesive matrix: coarse grains floating in a matrix of clay.

Flocculated structure has higher strength but is sensitive - when remoulded (disturbed) it breaks down and strength drops (sensitivity). Dispersed structure has lower permeability.

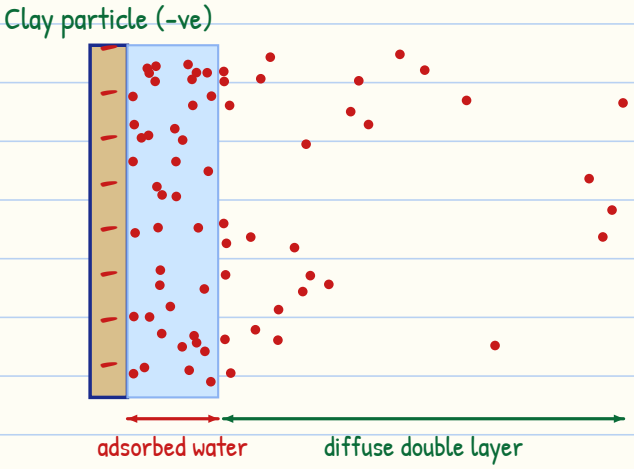
4. Clay Minerals

Clay minerals are hydrous aluminium silicates built from two basic units: the silica tetrahedron (Si) and the alumina octahedron (Al, gibbsite). Sheets stack to form layers.

Mineral	Structure	Bonding	Characteristics
Kaolinite	1:1 (Si + Al sheet)	H-bond (strong)	stable, low swelling, low plasticity
Illite	2:1, K ions between layers	K-bond (medium)	medium swelling and plasticity
Montmorillonite	2:1 (Al between 2 Si)	van der Waals (weak)	very high swelling and plasticity

- Isomorphous substitution: replacement of one cation by another of lower valency (e.g. Al^{3+} by Mg^{2+}) without changing crystal shape gives clay particles a net negative charge.
- Specific surface (surface area per unit mass): sand about 0.0001 m^2/g, kaolinite 10 - 20, illite 65 - 100, montmorillonite 700 - 840 m^2/g.
- Activity of clay = PI / (% clay fraction < 2 micron): inactive < 0.75, normal 0.75 - 1.25, active > 1.25 (montmorillonite about 7).

5. Adsorbed Water



+ve cations (red dots) are attracted to the negative clay surface; ion density falls with distance from the surface.

Water molecules are dipoles - they are held tightly near the surface (adsorbed water: very viscous, behaves like a solid).

Free / pore water lies beyond.

- Clay particles carry a negative charge on their surfaces and positive charge on their edges.
- Water molecules are dipoles (H^+ and O^{2-}); they are attracted to the clay surface by hydrogen bonds, by cations that are themselves hydrated, and by dipole attraction.
- Water held on the surface of the particle by these electro-chemical forces is called adsorbed water (also hygroscopic / bound water). It cannot be removed by gravity, and

extends to about 0.005 micron.

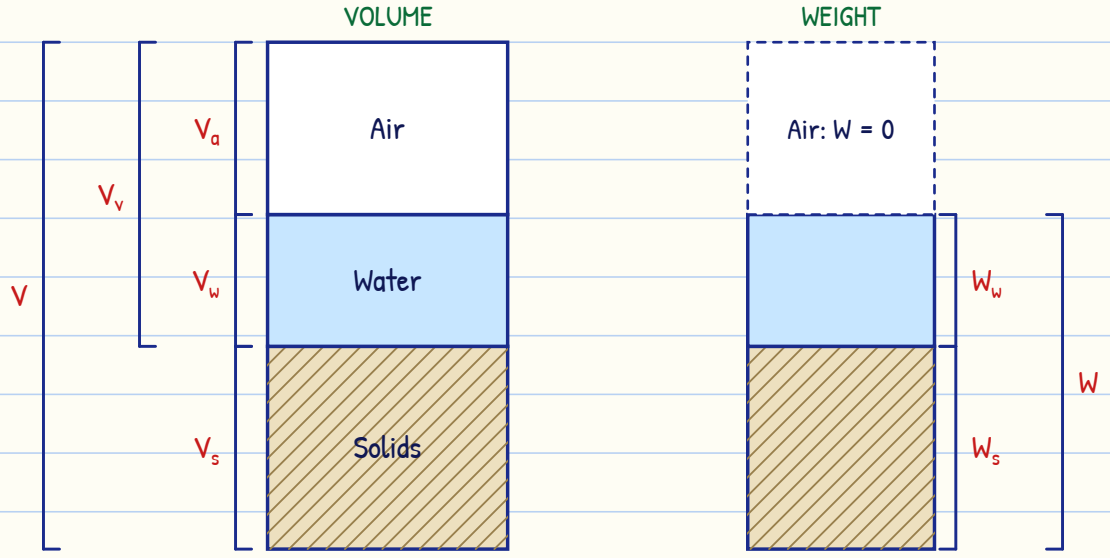
- The layer of adsorbed water + the cations around the particle is called the diffuse double layer. Thickness depends on valency, concentration of the electrolyte, and temperature.
- Thickness reduces when cation valency or concentration increases (e.g. Na^{+} gives thick, Ca^{2+} / Al^{3+} gives thin layer).
- Adsorbed water is much more viscous and has density a little higher than free water; it behaves like a solid and is removed only by oven drying at 105 - 110 degree C.

Effects of adsorbed water

- Gives plasticity and cohesion to clays (more adsorbed water, more plastic).
- Controls consistency limits, swelling, compressibility and permeability of clay.
- Makes clay structure flocculated or dispersed depending upon the double layer thickness.

6. Mass - Volume Relationships

Soil = solids + water + air. For analysis the three phases are shown separately in a phase diagram (three-phase diagram).



- V = total volume, V_s = volume of solids, V_w = volume of water, V_a = volume of air, V_v = $V_w + V_a$ = volume of voids.
- W = total weight, W_s = weight of solids, W_w = weight of water; weight of air is taken as zero. So $W = W_s + W_w$.

Volumetric ratios

- Void ratio $e = V_v / V_s$ (no unit; can be more than 1; sand 0.4 – 1.0, clay 0.5 – 1.5+).
- Porosity $n = V_v / V$ (expressed in %, always less than 100%).
- Degree of saturation $S = V_w / V_v$ (0 for dry soil, 100% for saturated soil).
- Air content $a_c = V_a / V_v$; so $a_c = 1 - S$.
- Percentage air voids $n_a = V_a / V$.

Weight ratios

- Water (moisture) content $w = W_w / W_s$ (can exceed 100%, e.g. peat, soft clays).
- Specific gravity of solids $G = (W_s/V_s) / \gamma_w$ (also called true or grain specific gravity).
- Mass specific gravity $G_m = \gamma / \gamma_w$ (bulk or apparent specific gravity).

Unit weights (densities)

Name	Symbol	Definition
Bulk (total) unit weight	γ	W / V
Dry unit weight	γ_d	W_s / V
Saturated unit weight	γ_{sat}	W / V when $S = 1$
Submerged (buoyant) unit weight	γ'	$\gamma_{sat} - \gamma_w$
Unit weight of solids	γ_s	$W_s / V_s = G \gamma_w$
Unit weight of water	γ_w	9.81 kN/m^3 (1 g/cc or 1000 kg/m^3)

Derivation of important relations (take $V_s = 1$, so $V_v = e$ and $V = 1 + e$)

- Relation between n and e: $n = V_v/V = e/(1+e)$ and $e = n/(1-n)$
- $W_s = G \gamma_w V_s = G \gamma_w$ and $W_w = w W_s = w G \gamma_w$
- $V_w = W_w/\gamma_w = w G$. But $S = V_w/V_v = wG/e$.

$$S e = w G$$

- Dry unit weight $\gamma_d = W_s/V = G \gamma_w / (1 + e)$
- Bulk unit weight $\gamma = (W_s + W_w)/V = (1 + w) G \gamma_w / (1 + e) = (G + S e) \gamma_w / (1 + e)$

$$\gamma_d = \gamma / (1 + w)$$

- Saturated ($S = 1$): $\gamma_{sat} = (G + e) \gamma_w / (1 + e)$
- Submerged: $\gamma' = \gamma_{sat} - \gamma_w = (G - 1) \gamma_w / (1 + e)$
- Dry soil ($S = 0$): $\gamma = \gamma_d = G \gamma_w / (1 + e)$
- Air voids: $n_a = n (1 - S) = (1 - S) e / (1+e)$
- Zero air voids line ($S = 1$): $\gamma_d = G \gamma_w / (1 + w G)$

Order of unit weights: $\gamma_{sat} > \gamma > \gamma_d > \gamma'$. Also $e = (G \gamma_w / \gamma_d) - 1$.

Solved example 1

A soil has bulk density 1.92 g/cc, water content 12% and G = 2.70. Find dry density, void ratio, porosity and degree of saturation.

- $\gamma_d = \gamma/(1 + w) = 1.92/1.12 = 1.714$ g/cc
- $e = G\gamma_w/\gamma_d - 1 = 2.70/1.714 - 1 = 0.575$
- $n = e/(1+e) = 0.575/1.575 = 0.365 = 36.5$ %
- $S = wG/e = (0.12 \times 2.70)/0.575 = 0.563 = 56.3$ %

Solved example 2

A fully saturated clay has w = 45% and G = 2.72. Find e, γ_{sat} and γ'.

- S = 1 so $e = wG = 0.45 \times 2.72 = 1.224$
- $\gamma_{sat} = (G + e)\gamma_w/(1 + e) = (2.72 + 1.224) \times 9.81/2.224 = 17.40$ kN/m^3
- $\gamma' = 17.40 - 9.81 = 7.59$ kN/m^3

7. Relative Density (Density Index)

Used for cohesionless soils to say how dense a sand is in its natural state compared with its loosest and densest states.

$$D_r = (e_{max} - e) / (e_{max} - e_{min}) \times 100\ \%$$

- Also $D_r = [\ \gamma_{d\,max}(\gamma_d - \gamma_{d\,min})\ ] / [\ \gamma_d(\gamma_{d\,max} - \gamma_{d\,min})\ ] \times 100$
- e_{max} = void ratio in loosest state (poured through funnel), e_{min} = void ratio in densest state (vibration + surcharge), e = in-situ void ratio.
- D_r = 0 for the loosest and D_r = 100 % for the densest state.

Relative density (%)	0 - 15	15 - 35	35 - 65	65 - 85	85 - 100
Description	Very loose	Loose	Medium dense	Dense	Very dense

Example

Sand: $e_{max} = 0.85$, $e_{min} = 0.40$, natural e = 0.62.

$D_r = (0.85 - 0.62)/(0.85 - 0.40) = 0.23/0.45 = 0.511 = 51.1$ % - medium dense sand.

Relative density is related to Standard Penetration Test N-value: N < 4 very loose, 4-10 loose, 10-30 medium, 30-50 dense, > 50 very dense. (Terzaghi and Peck)

8. Index Properties of Soils

Index properties are simple tests on disturbed samples used to identify and classify soils and to correlate their engineering behaviour (strength, compressibility, permeability). Two types: soil grain properties (grain size, shape, specific gravity) and soil aggregate properties (water content, consistency, density, relative density).

8.1 Moisture (water) content - w

$$w = W_w / W_s \times 100 = (W_2 - W_3) / (W_3 - W_1) \times 100$$

- Oven drying method (IS 2720 Part 2): W_1 = wt of empty container, W_2 = container + wet soil, W_3 = container + oven dry soil. Dry at 105 - 110 degree C for 24 hours (till constant mass). Most accurate, standard method.
- Pycnometer method: quick, for coarse-grained soil; $w = [(W_2 - W_1) / (W_3 - W_4) \times (G - 1)/G - 1] \times 100$.
- Calcium carbide (Speedy moisture meter): soil + CaC_2 reacts with water giving acetylene gas, whose pressure on the gauge gives moisture. Field use, 5 min.
- Others: sand bath, alcohol method, infrared lamp, torsion balance, radio-isotope method.

8.2 Specific gravity of soil solids - G

$$G = (W_2 - W_1) / [(W_2 - W_1) - (W_3 - W_4)]$$

- Density bottle / pycnometer method (IS 2720 Part 3): W_1 = empty bottle, W_2 = bottle + dry soil, W_3 = bottle + soil + water, W_4 = bottle + water only.
- Procedure: weigh the dry bottle, add 200 g (or 10 - 25 g in a density bottle) of oven-dry soil, add water, remove entrapped air by vacuum / boiling, fill to mark, weigh, then fill with water only and weigh.
- For soils with soluble salts use kerosene instead of water.
- Gas jar method for coarse grained (> 4.75 mm) soils; Pycnometer for fine and medium grained soils.

Soil	Sand	Silt	Clay	Organic soil	Laterite	Gravel
G	2.65 - 2.67	2.67 - 2.70	2.70 - 2.80	1.4 - 2.3	2.75 - 3.0	2.65 - 2.68

- Use of G: needed for e, S, γ_d, hydrometer analysis, zero-air-void line and compaction.

8.3 In-situ (field) density

Determines γ and γ_d of soil in place; used for compaction control and bearing capacity.

$$\gamma = W / V \qquad \gamma_d = \gamma / (1 + w)$$

- Core cutter method (IS 2720 Part 29): for soft to medium cohesive soils (no gravel). Standard cutter: 100 mm dia, 130 mm height, volume about 981 cc. Drive the cutter, remove with soil, trim ends, weigh: $\gamma = (W_2 - W_1) / V_c$.
- Sand replacement method (IS 2720 Part 28): for gravelly and sandy soils. A hole of about 100 mm dia is dug, soil weighed (W), then the hole is filled with calibrated sand from a pouring cylinder. $V_{hole} = W_{sand\ in\ hole} / \gamma_{sand}$, where $W_{sand\ in\ hole} = W_1 - W_2 - W_{cone}$. Then $\gamma = W / V_{hole}$.
- Water displacement / wax method: for irregular lumps, soil coated with paraffin wax and weighed in air and water.
- Nuclear density gauge / rubber balloon method: quick and non-destructive.

Solved example 3 (sand replacement)

Excavated soil = 1.85 kg, water content 10 %. Sand filling the hole = 1.30 kg, density of sand = 1.45 g/cc. Find γ and γ_d.

- $V_{hole} = 1300/1.45 = 896.6$ cc; $\gamma = 1850/896.6 = 2.063$ g/cc
- $\gamma_d = 2.063/1.10 = 1.876$ g/cc

9. Grain Size Analysis (Mechanical Analysis)

Determines the relative proportion of different grain sizes in a soil, plotted as the particle size distribution curve.

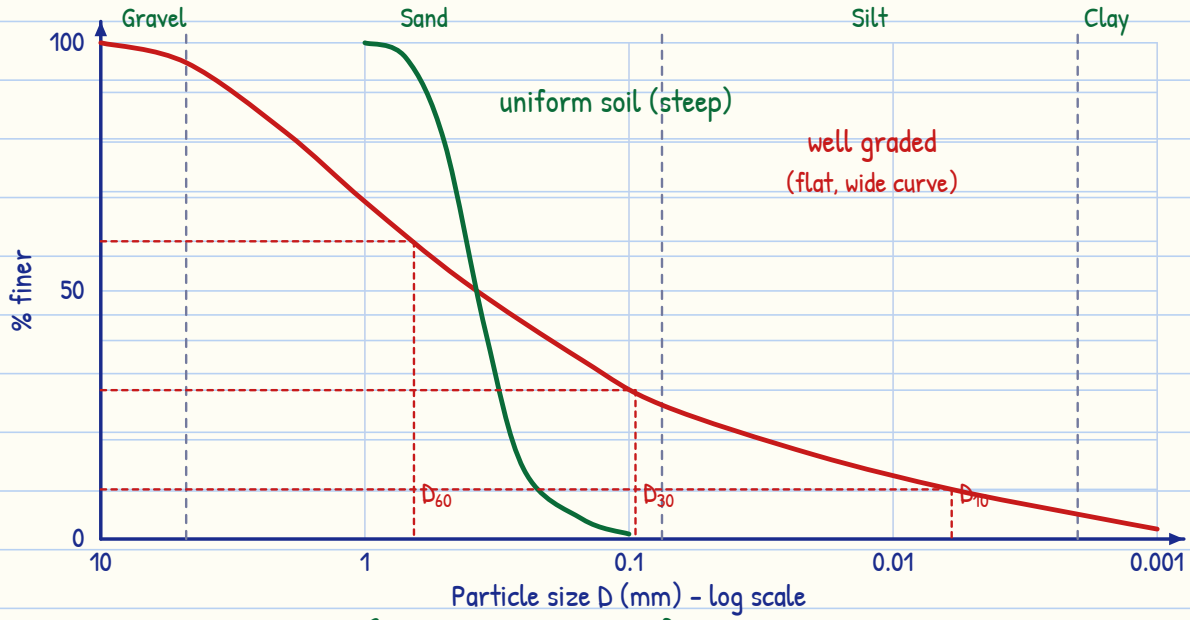
Soil fraction	Gravel	Coarse sand	Medium sand	Fine sand	Silt	Clay
Size (IS)	80 - 4.75 mm	4.75 - 2.0 mm	2.0 - 0.425 mm	0.425 - 0.075 mm	0.075 - 0.002 mm	< 0.002 mm

9.1 Sieve analysis (for particles > 75 micron)

- Used for coarse-grained soils. A set of IS sieves is arranged in descending order of opening: 4.75 mm, 2.0 mm, 1.0 mm, 600 micron, 425 micron, 300 micron, 212 micron, 150 micron, 75 micron, then the pan.
- Oven-dry soil (500 g - 1 kg) is washed over the 75 micron sieve (wet sieving for soils with fines), dried and then shaken for 10 minutes in a mechanical sieve shaker.
- Weight retained on each sieve is found.

% retained = (W retained on sieve / W total) x 100; % finer N = 100 - cumulative % retained

- Plot % finer (arithmetic) vs particle size D (log scale).



Grading parameters (from the curve)

- Effective size D_{10}: size such that 10 % of soil is finer. D_{30} and D_{60} similarly.

$$C_u = D_{60} / D_{10} \qquad C_c = (D_{30})^2 / (D_{10} \times D_{60})$$

- Well graded: gravel $C_u > 4$, sand $C_u > 6$ AND $1 < C_c < 3$ (all sizes present, good compaction).
- Poorly graded (uniform): does not satisfy the above; gap graded has some sizes missing (hump in curve).
- Steep curve means uniform soil, flat curve means well graded.

Example

D_{10} = 0.10 mm, D_{30} = 0.30 mm, D_{60} = 0.80 mm: C_u = 0.80/0.10 = 8; C_c = $0.30^2/(0.10 \times 0.80)$ = 1.125. $C_u > 6$ and C_c between 1 and 3, so it is a well graded sand.

9.2 Hydrometer analysis (for particles < 75 micron)

- Based on Stokes' law: the terminal velocity of a spherical particle settling in a liquid is $v = (\gamma_s - \gamma_w) D^2 / (18 \eta)$.

$$D = \text{sqrt}\left[18 \eta / (\gamma_s - \gamma_w) \times (H_e / t) \right] \quad (D \text{ in mm, } t \text{ in min})$$

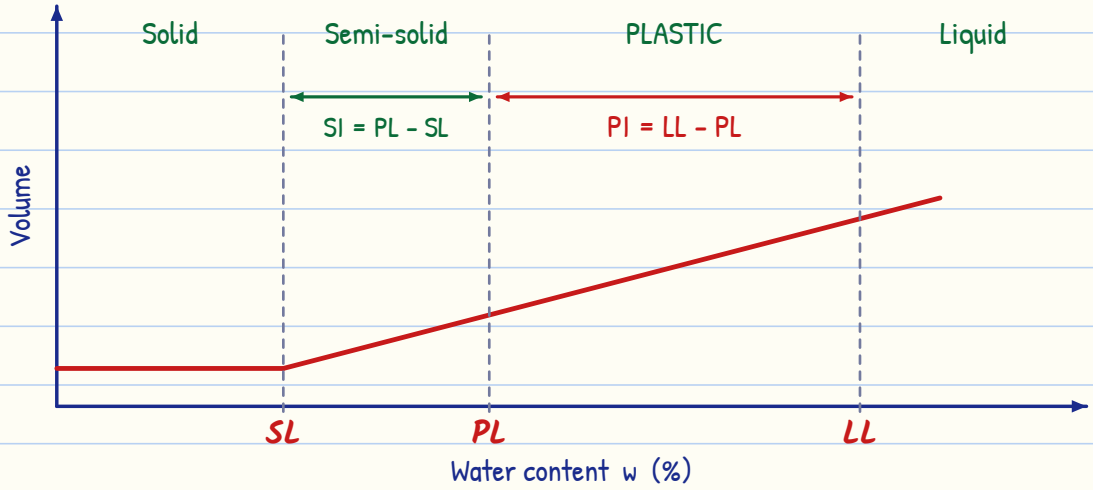
- η = viscosity of water, H_e = effective depth (from the surface to the centre of the hydrometer bulb), t = time of settling.
- Soil (50 g) + dispersing agent (sodium hexametaphosphate or sodium silicate, 2 %) is made into a 1000 ml suspension. Hydrometer readings (152H) are taken at 0.5, 1, 2, 4, 8, 15, 30, 60, 250 and 1440 minutes.
- Corrections: meniscus correction (C_m, always added), dispersing agent correction (C_d, subtracted) and temperature correction (C_t, + or -).

- % finer $N = (R_c \times G) / [\, W_s (G - 1)\,]$, where R_c is the corrected hydrometer reading.
- The results are combined with the sieve analysis result to obtain the complete grain size curve.
- Pipette method is more accurate but is a laboratory method; samples are drawn from a fixed depth with a pipette.

Stokes' law assumptions: particles are spherical, smooth, settle independently with no interference, and the velocity is small (laminar). Valid for 0.2 mm > D > 0.2 micron only.

10. Consistency Limits (Atterberg Limits)

A fine-grained soil can exist in four states depending on water content: solid, semi-solid, plastic and liquid. The boundary water contents were suggested by Atterberg (1911) and standardised by Casagrande.



Liquid limit w_L (LL)

Water content at which the soil changes from plastic to liquid state; it has a very small shear strength (about 2.7 kPa).

- Casagrande's apparatus (IS 2720 Part 5): soil paste (passing 425 micron) is placed in the brass cup, a groove is cut with a standard grooving tool and the cup is dropped through 1 cm at 2 drops/sec. LL = water content at which the groove closes by 12 mm in 25 blows.
- Take 3 - 4 trials with number of blows from 10 to 40 and plot flow curve (w vs log N). Flow index $I_f = (w_1 - w_2) / \log (N_2/N_1)$ = slope of flow curve.

- Cone penetration method: 30 degree cone weighing 148 g is released on the soil in a cup; LL is the water content corresponding to 25 mm penetration in 5 seconds.

Plastic limit w_P (PL)

Water content at which soil changes from semi-solid to plastic state. The soil is rolled by hand on a glass plate into a thread; PL = the water content at which the thread of 3 mm diameter just begins to crumble.

Shrinkage limit w_S (SL)

Water content below which there is no further decrease in volume of the soil on drying.

$$w_S = w_1 - [\,(V_1 - V_2) / W_d\,]\,\gamma_w \times 100$$

- w_1 = initial water content, V_1 = initial volume of wet pat, V_2 = volume of dry pat (by mercury displacement), W_d = dry weight.

Consistency indices

Index	Formula	Meaning
Plasticity index PI	$w_L - w_P$	range of water content over which soil is plastic
Shrinkage index SI	$w_P - w_S$	range of semi-solid state
Liquidity index LI	$(w - w_P) / PI$	LI < 0 stiff, 0 – 1 plastic, > 1 liquid
Consistency index CI	$(w_L - w) / PI$	CI = 1 – LI
Flow index I_f	slope of flow curve	rate of change of w with log N
Toughness index I_t	PI / I_f	shear strength at plastic limit
Activity A	PI / % clay (< 2 micron)	swelling potential

- Sensitivity S_t = (unconfined strength undisturbed) / (strength remoulded).
- Degree of plasticity (IS): PI 0 non-plastic, 1 – 7 low, 7 – 17 medium, 17 – 35 high, > 35 very high.

Solved example 4

LL = 48 %, PL = 25 %, natural water content = 36 %. Find PI, LI, CI and state of soil.

- PI = 48 – 25 = 23 %
- LI = (36 – 25)/23 = 0.478; CI = (48 – 36)/23 = 0.522
- 0 < LI < 1, so the soil is in the plastic state (medium stiff).

Sand and silt: PI = 0 (non-plastic). Higher PI means more clay content and higher compressibility and swelling.

II. I.S. Classification of Soils (IS 1498 - 1970)

Soils are grouped by particle size (coarse) and plasticity (fines). Three main divisions:

Division	Criterion	Groups
Coarse-grained	more than 50 % retained on 75 micron IS sieve	Gravel (G) and Sand (S)
Fine-grained	more than 50 % passes 75 micron IS sieve	Silt (M), Clay (C), Organic (O)
Highly organic	peat and other fibrous soils	Pt

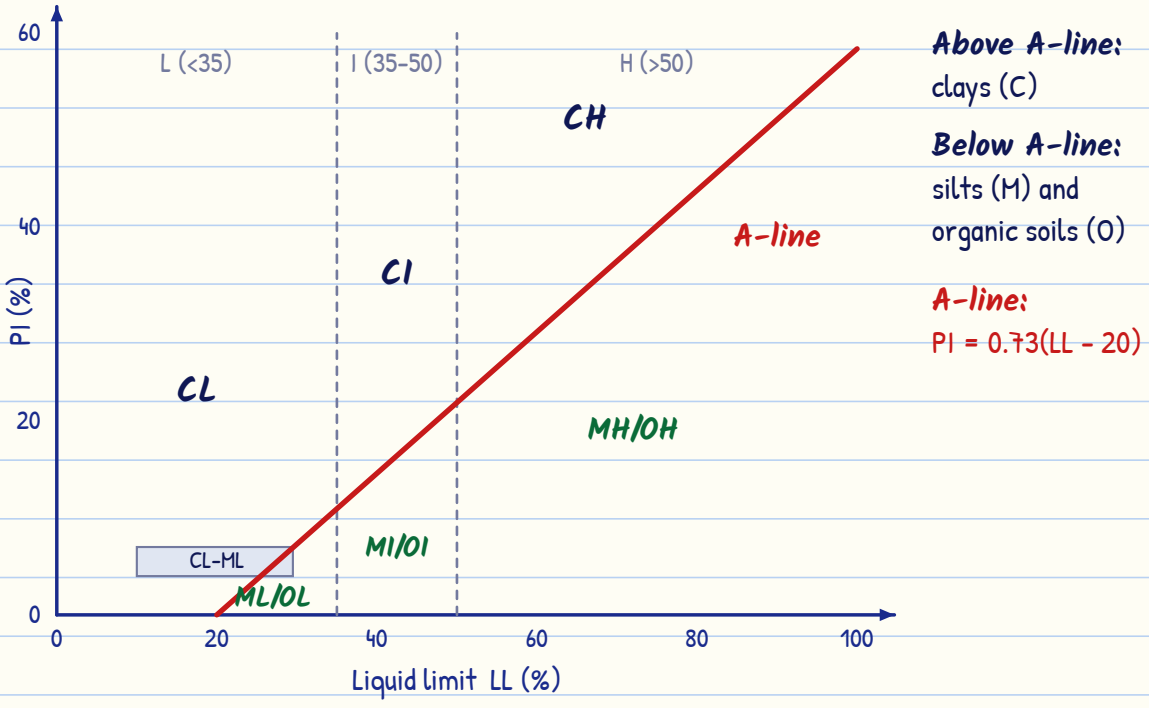
Coarse-grained soils

- Gravel (G): more than 50 % of the coarse fraction is retained on the 4.75 mm sieve. Sand (S): more than 50 % of coarse fraction passes 4.75 mm.
- Fines (passing 75 micron) < 5 %: W (well graded) or P (poorly graded) -> GW, GP, SW, SP.
- Fines > 12 %: M or C according to plasticity -> GM, GC, SM, SC.
- Fines 5 - 12 %: dual symbols e.g. GW-GM, SP-SC.

Group	GW	GP	SW	SP
Condition	$C_u > 4$ and $1 < C_c < 3$	not GW	$C_u > 6$ and $1 < C_c < 3$	not SW

Fine-grained soils (use the plasticity chart)

- LL < 35 %: L (low compressibility), 35 - 50 %: I (intermediate), LL > 50 %: H (high).
- Above the A-line (PI = 0.73 (LL - 20)): inorganic clays CL, CI, CH.
- Below the A-line: inorganic silts ML, MI, MH and organic silts / clays OL, OI, OH.
- CL-ML: hatched zone (PI between 4 and 7) near the A-line; also called borderline soils.



Group symbols – summary

Symbol	Meaning	Symbol	Meaning
GW	well graded gravel	GM	silty gravel
GP	poorly graded gravel	GC	clayey gravel
SW	well graded sand	SM	silty sand
SP	poorly graded sand	SC	clayey sand
ML	silt of low compress.	CL	clay of low plasticity
MI	silt, intermediate	CI	clay, intermediate
MH	silt of high compress.	CH	clay of high plasticity
OL/OI/OH	organic silts and clays	Pt	peat

Field identification of soils (quick tests)

- Dry strength, dilatancy (shaking test), toughness (thread test) and visual inspection (colour, odour) classify fine soils in the field.
- Silts show quick dilatancy and low dry strength; clays show no dilatancy and high dry strength; organic soils have a spongy feel and odour.

12. Quick Revision – Important Formulas

Quantity	Formula
Void ratio / porosity	$e = V_v/V_s$; $n = e/(1+e)$
S e = w G	Dry unit weight $\gamma_d = G\gamma_w/(1+e) = \gamma/(1+w)$
Saturated unit weight	$\gamma_{sat} = (G + e)\gamma_w/(1+e)$
Submerged unit weight	$\gamma' = (G - 1)\gamma_w/(1+e)$
Relative density	$D_r = (e_{max} - e)/(e_{max} - e_{min})$
Uniformity coefficient	$C_u = D_{60}/D_{10}$
Coefficient of curvature	$C_c = D_{30}^2/(D_{10} D_{60})$
Plasticity index	PI = LL – PL
Liquidity index	LI = (w – PL)/PI
A-line	PI = 0.73 (LL – 20)

Important questions (exam)

- Explain the formation of soils and the different types of soil structures.
- Define void ratio, porosity, degree of saturation and derive e, S, w, G relation: S e = w G.

- Derive γ_{sat}, γ_d and γ' in terms of G and e.
- Explain the sand replacement method for finding in-situ density.
- Describe the hydrometer analysis with Stokes' law.
- Explain Atterberg limits and the indices of consistency; how are they determined?
- Explain the IS classification of soil and the plasticity chart.

GEOTECHNICAL ENGINEERING

UNIT - II : PERMEABILITY & SEEPAGE THROUGH SOILS

Course Code: 23A01503T

Syllabus: PERMEABILITY: Soil water - Capillary rise - Flow of water through soils - Darcy's law - Permeability, factors affecting - Laboratory determination of coefficient of permeability - Permeability of layered systems. SEEPAGE THROUGH SOILS: Total, neutral and effective stresses - Quick sand condition - Seepage through soils - Flow nets: characteristics and uses.

1. Soil Water

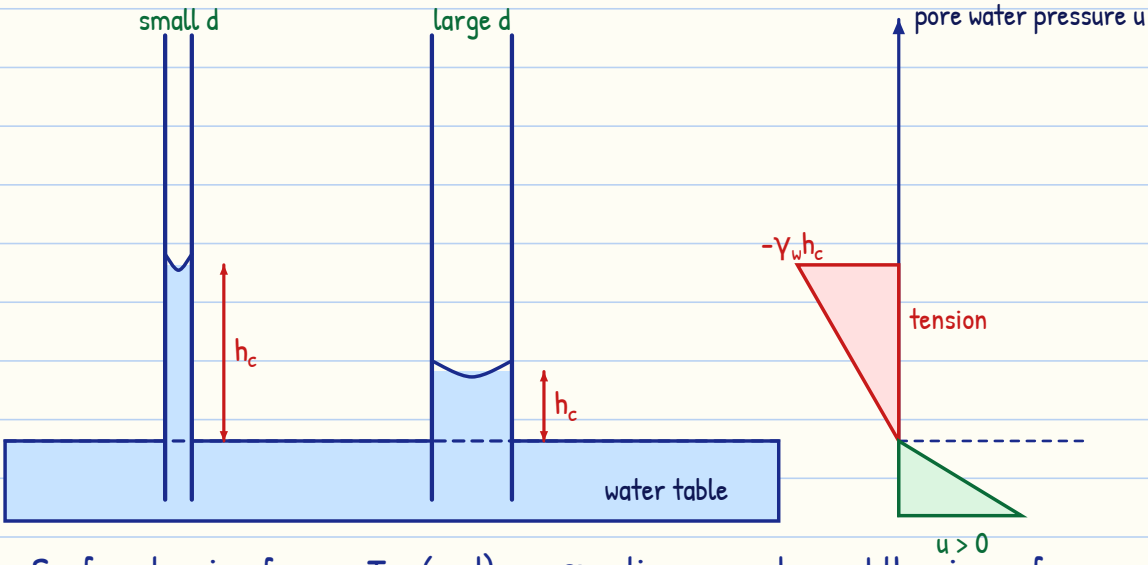
Water in the soil can be classified according to how it is held in the voids and on the grains.

Type	Description
Free (gravitational) water	Moves under gravity, drainage; found below the water table
Capillary water	Held in the fine voids above water table by surface tension
Adsorbed (hygroscopic) water	Held on clay surfaces by electro-chemical forces; not removable by gravity
Structural (crystal) water	Part of the crystal lattice of minerals, removed only at very high temperatures

- Ground water table (WT): the surface at which the pore water pressure is equal to atmospheric ($u = 0$). Below it, soil is saturated.
- Phreatic surface: the top flow line of seepage in an earth dam; same idea as the WT.
- Perched water table: local water table above an impermeable lens, above the main water table.
- Artesian condition: water in a confined aquifer under pressure greater than atmospheric (head above the ground level).

2. Capillary Rise

Water rises above the water table in a fine tube (or in the fine pores of soil) due to surface tension T of water and the adhesion to the tube walls.



- Surface tension forces: $T \times (\pi\ d) \cos\alpha$ acting upward round the circumference. Weight of the lifted water column: $(\pi\ d^2/4)\ h_c\ \gamma_w$.

$$h_c = 4\ T \cos\alpha\ /\ (\gamma_w\ d)$$

- T = surface tension of water = 0.073 N/m (at 20 degree C); α = contact angle (0 degree for clean glass and water); d = diameter of tube. For water: h_c (cm) = 0.3 / d (cm).
- In soils the equivalent pore diameter is about $D_{10}/5$, so: $h_c = C / (e\ D_{10})$, where C = 0.1 to 0.5 cm^2 (Hazen / Terzaghi-Peck).

Points to remember

- Smaller the pore (finer the soil), higher the capillary rise, but the rate of rise is slower.
- Above the water table, the pore water pressure is negative (tension): $u = -\gamma_w\ h$ at height h above WT. At the top of the capillary fringe $u = -\gamma_w\ h_c$.
- Capillary fringe may be saturated or partially saturated; fully saturated zone (about 2/3 h_c) then partially saturated zone above.

Soil	Coarse sand	Fine sand	Silt	Clay
Approx. capillary rise	0.03 - 0.12 m	0.3 - 3.5 m	1.5 - 12 m	2 - 50 m

Effects / importance of capillarity

- Negative pore pressure raises the effective stress, giving apparent cohesion (damp sand holds a sand castle; dry or fully submerged sand has none).
- Bulking of sand: moist sand (water content 4 - 6 %) occupies up to 20 - 30 % more volume, since capillary films keep the grains apart.
- Frost heave: capillary water feeds ice lenses and heaves pavements in cold regions.
- Capillary water makes foundations, basements and road subgrades damp (damp-proof course is needed); shrinkage of clays is due to capillary tension.

Example

For a soil, D_{10} = 0.02 mm = 0.002 cm, e = 0.60, C = 0.3 cm^2.

h_c = 0.3 / (0.60 x 0.002) = 250 cm = 2.5 m.

3. Flow of Water Through Soils

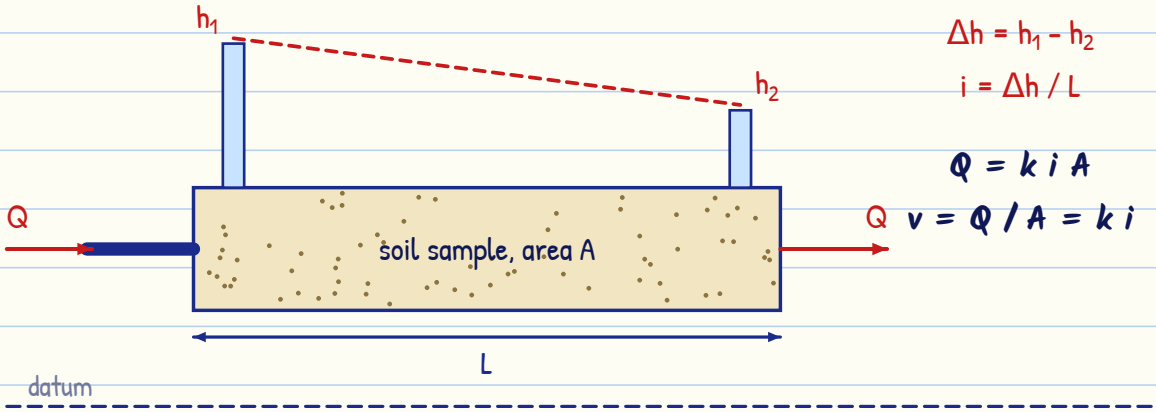
Water flows through the interconnected voids. The driving force is the difference in total head between two points.

- By Bernoulli's equation the total head $h = (u / \gamma_w) + z + (v^2 / 2g)$. In soils the velocity v is very small, so the velocity head is neglected:

$$h = u / \gamma_w + z \quad \text{(pressure head + elevation head)}$$

- Water always flows from higher total head to lower total head, not necessarily from high pressure to low pressure.
- Hydraulic gradient $i = \Delta h / L$ = head loss per unit length of flow path (dimensionless).
- Flow is laminar in soils (Reynolds number $Re = v D \rho / \eta < 1$, in most soils; it can be turbulent in coarse gravel / rock fill).

Darcy's Law (1856)



Henry Darcy, studying water filters at Dijon (France), found experimentally that for laminar flow through saturated soil the discharge velocity is directly proportional to hydraulic gradient.

$$v = k i \qquad q = Q = k i A$$

- v = discharge (superficial) velocity = Q/A; k = coefficient of permeability (hydraulic conductivity) with the dimensions of velocity (cm/s, m/s, m/day); A = total cross-section area of soil (solids + voids).

- Coefficient of permeability k: the discharge velocity under unit hydraulic gradient (i = 1).
- The real velocity in the pores is the seepage velocity v_s. Flow area is only the void area $A_v = n A$, therefore

$$v_s = v / n = v (1 + e) / e \qquad (v_s > v)$$

Validity of Darcy's law

- Valid for laminar flow (Re < 1; up to 10 for practical use), saturated soil, steady flow, incompressible soil and water.
- Not valid in very coarse gravels / rock fill (turbulent), and in very fine clays at very low gradients (below a threshold gradient).

Typical values of k

Soil	Clean gravel	Coarse sand	Fine sand	Silty sand	Silt	Clay
k (cm/s)	> 1	$1 - 10^{-2}$	$10^{-2} - 10^{-3}$	$10^{-3} - 10^{-5}$	$10^{-5} - 10^{-7}$	$< 10^{-7}$
Drainage	very good	good	good	poor	poor	practically impervious

k of clean sand is about a million times that of clay. Remember: k is a property of both the soil and the fluid. $k = K \gamma_w / \eta$, where K = intrinsic (absolute) permeability (cm^2).

4. Factors Affecting Permeability

- Grain size: k increases with the square of the particle size. Hazen: $k = C D_{10}^2$ (k in cm/s, D_{10} in mm, C = 100 for loose clean sand, range 100 - 150).
- Void ratio: larger e gives larger k. Different relations: k proportional to e^2, or $e^3/(1+e)$ (Kozeny-Carman). For sands $k_1 / k_2 = [e_1^3/(1+e_1)] / [e_2^3/(1+e_2)]$.
- Shape and texture of the particles: angular grains give a more tortuous flow path and lower k than rounded grains; rough surface lowers k.
- Structure and stratification: flocculated clays have higher k than dispersed; k parallel to the bedding is more than k perpendicular ($k_h > k_v$).
- Degree of saturation: partly saturated soil (air bubbles block pores) has lower k, so use fully saturated, de-aired water in tests.
- Properties of the permeant: k is proportional to γ_w / η. Viscosity falls as temperature rises so k increases with temperature. Results are reported at 27 degree C (IS) or 20 degree C.

$$k_{27} = k_T \times (\eta_T / \eta_{27})$$

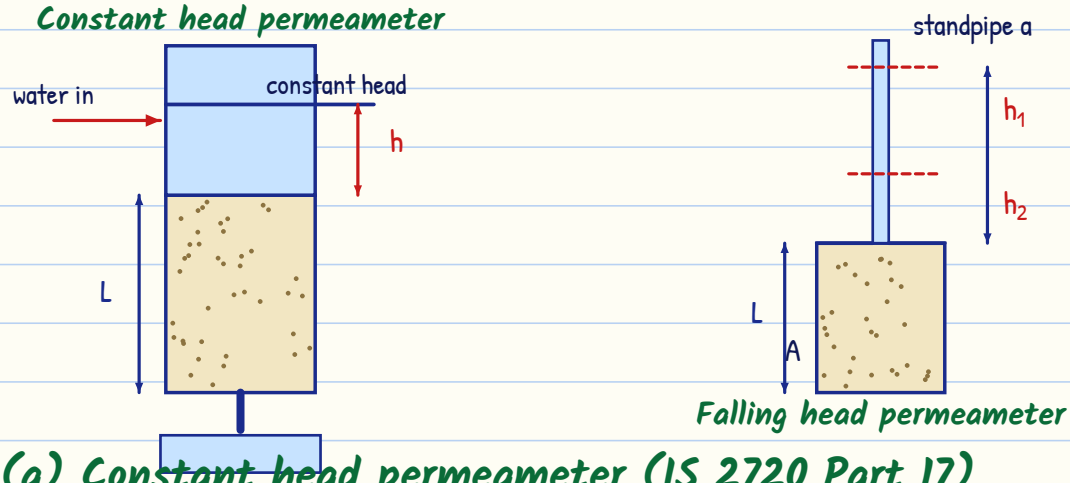
- Adsorbed water in clays blocks the pore throat, lowering k. Impurities / organic matter / entrapped air also reduce k.
- Casagrande: for fine sands, $k = 1.4\, e^2\, k_{0.85}$ ($k_{0.85}$ is k at e = 0.85).

Example (effect of void ratio)

For a sand, $k = 4 \times 10^{-2}$ cm/s at e = 0.60. Find k at e = 0.80.

$k_2 = k_1 \times [e_2^3/(1+e_2)] / [e_1^3/(1+e_1)] = 4 \times 10^{-2} \times (0.512/1.8) / (0.216/1.6) = 4 \times 10^{-2} \times 0.2844 / 0.135 = 8.4 \times 10^{-2}$ cm/s.

5. Laboratory Determination of Coefficient of Permeabi



(a) Constant head permeameter (IS 2720 Part 17)

Suitable for coarse-grained soils (clean sand and gravel) having $k > 10^{-3}$ cm/s. Head h is maintained constant, so the discharge Q is steady.

- Soil sample of length L and cross-sectional area A is placed in a cylindrical mould (100 mm dia, 127.3 mm long); water is passed under constant head h, and the quantity Q collected in time t is measured.

$$k = Q\,L / (A\,h\,t)$$

- With a constant head, the water collected in time t equals $Q = k\,(h/L)\,A\,t$.

Example 1

A sample 15 cm long and 50 cm^2 in area under a constant head of 40 cm gave 450 cc of water in 2 minutes. Find k.

$k = (450 \times 15) / (50 \times 40 \times 120) = 6750 / 240000 = 0.028$ cm/s.

(b) Falling head permeameter (IS 2720 Part 17)

Suitable for fine-grained soils (silt and clay), $k < 10^{-3}$ cm/s. Water flows from a standpipe of area a through the sample; head falls from h_1 to h_2 in time t.

- At any instant, rate of fall: $-a\,dh/dt = k\,(h/L)\,A$. Integrating from h_1 to h_2:

$$k = (a\,L / A\,t)\,\ln(h_1 / h_2) = 2.303\,(a\,L / A\,t)\,\log_{10}(h_1 / h_2)$$

Example 2

Falling head test: $a = 0.5$ cm^2, $A = 30$ cm^2, $L = 12$ cm, $h_1 = 100$ cm, $h_2 = 60$ cm, $t = 10$ minutes.

$k = [0.5 \times 12 / (30 \times 600)] \times \ln(100/60) = 3.33 \times 10^{-4} \times 0.5108 = 1.7 \times 10^{-4}$ cm/s.

Comparison

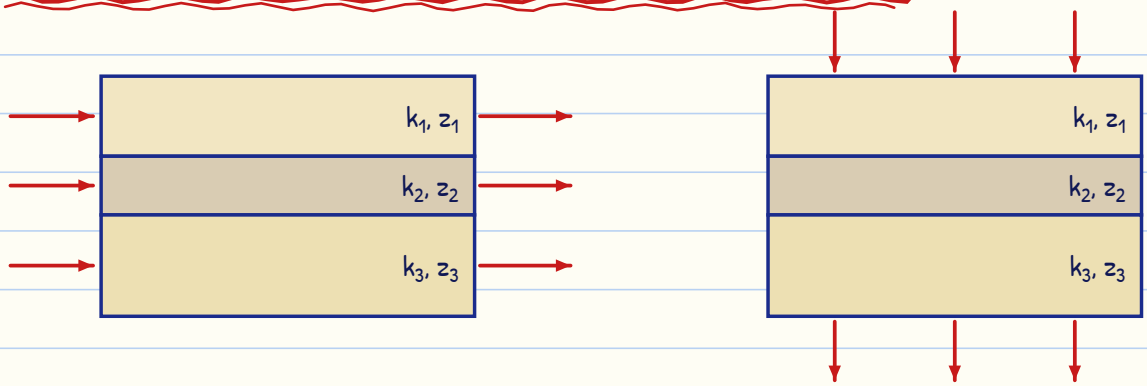
Constant head	Falling head
coarse soils, $k > 10^{-3}$	fine soils, $k < 10^{-3}$
head constant, Q measured	head falls, time measured
quick, reliable for sands	accurate for silts / clays

(c) Other methods

- Indirect methods: from grain size (Hazen's formula), from the consolidation test ($k = c_v\,m_v\,\gamma_w$) and from capillarity test (horizontal).
- Field methods: pumping-out tests (for unconfined aquifer $k = Q\ln(r_2/r_1) / [\pi\,(h_2^2 - h_1^2)]$; for confined aquifer $k = Q\ln(r_2/r_1) / [2\pi\,D\,(h_2 - h_1)]$), borehole tests, tracer / slug tests.
- Field tests give more representative value as they include layering and fissures.

In a laboratory test the sample should be fully saturated and flow upward so that air is carried out; keep the hydraulic gradient low to avoid piping in loose sand.

6. Permeability of Layered Systems



Flow parallel to layers *Flow perpendicular to layers*

Natural deposits are stratified, so the equivalent permeability depends on the direction of flow. Let layers have permeabilities $k_1, k_2, \ldots k_n$ and thicknesses $z_1, z_2, \ldots z_n$ with total thickness $Z = \Sigma z$.

(i) Flow parallel to the layers (horizontal flow)

- The hydraulic gradient i is the same in all layers; total discharge = sum of discharges.

$$k_h = (k_1 z_1 + k_2 z_2 + \ldots + k_n z_n) / (z_1 + z_2 + \ldots + z_n)$$

(ii) Flow perpendicular to the layers (vertical flow)

- The discharge velocity is the same in all layers; total head loss = sum of head losses in the layers.

$$k_v = (z_1 + z_2 + \ldots + z_n) / (z_1/k_1 + z_2/k_2 + \ldots + z_n/k_n)$$

- k_h is controlled by the most permeable layer; k_v is controlled by the least permeable layer. Always $k_h > k_v$; for natural clay deposits k_h/k_v = 2 to 10, and for varved clay up to 20 or more.
- For a single homogeneous but anisotropic soil, the equivalent isotropic permeability k_e = sqrt (k_h k_v).

Example 3

Three horizontal layers: ($k_1 = 2 \times 10^{-4}$ cm/s, z_1 = 2 m), ($k_2 = 5 \times 10^{-5}$, z_2 = 3 m), ($k_3 = 1 \times 10^{-3}$, z_3 = 1 m). Find k_h and k_v.

- $k_h = (2 \times 10^{-4} \times 2 + 5 \times 10^{-5} \times 3 + 10^{-3} \times 1) / 6 = 1.55 \times 10^{-3} / 6 = 2.58 \times 10^{-4}$ cm/s
- $k_v = 6 / (2/2 \times 10^{-4} + 3/5 \times 10^{-5} + 1/10^{-3}) = 6 / (10000 + 60000 + 1000) = 8.45 \times 10^{-5}$ cm/s
- Ratio k_h/k_v = 3.05.

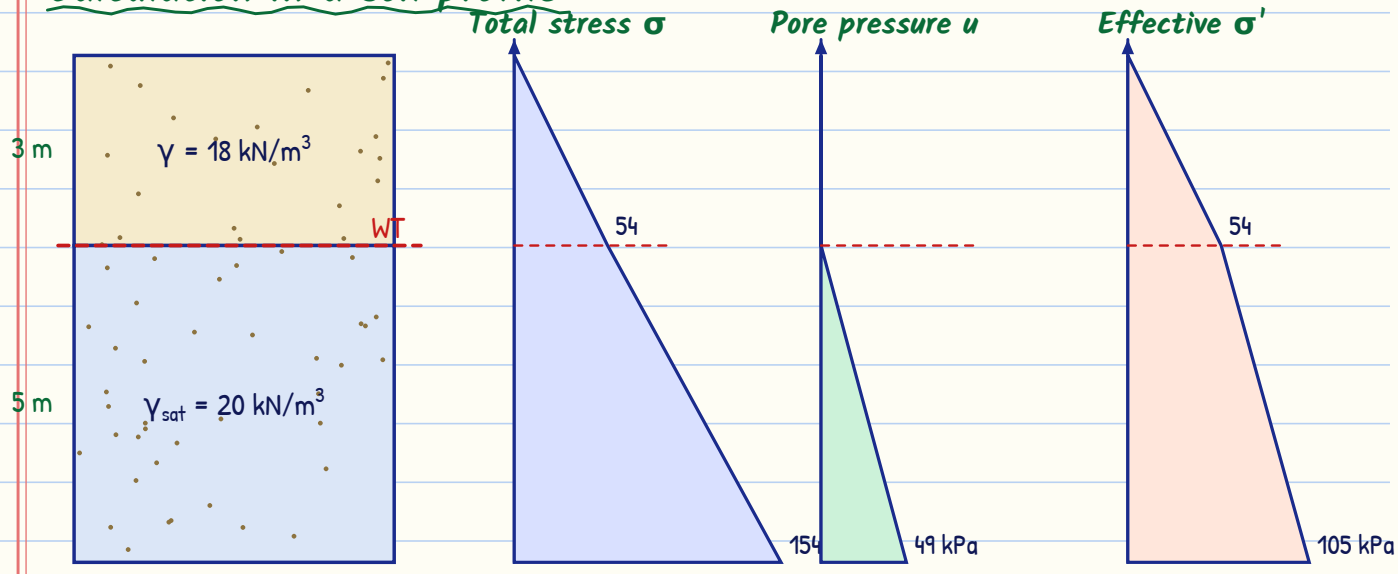
7. Total, Neutral and Effective Stresses

Terzaghi (1923) - the principle of effective stress: in a saturated soil, the total stress σ at a point is shared by the soil skeleton (effective stress σ') and the pore water (neutral stress or pore pressure u).

$$\sigma = \sigma' + u \qquad \text{or} \qquad \sigma' = \sigma - u$$

- Total stress σ = weight of everything (soil + water) above the point per unit area.
- Neutral stress u (pore water pressure) acts equally in all directions; it cannot cause deformation or shear strength. $u = \gamma_w h_w$.
- Effective stress σ' = the intergranular stress carried through the grain-to-grain contacts. All measurable effects - compression, distortion and shear strength - depend upon the effective stress, not the total stress.

Calculation in a soil profile



Sand (γ = 18 kN/m³) from 0 - 3 m; water table at 3 m; clay (γ_{sat} = 20 kN/m³) from 3 - 8 m. At 8 m depth:

- $\sigma = 18 \times 3 + 20 \times 5 = 54 + 100 = 154$ kPa
- $u = 9.81 \times 5 = 49.05$ kPa
- $\sigma' = 154 - 49.05 = 104.95$ kPa (also $\sigma' = 18 \times 3 + (20 - 9.81) \times 5 = 54 + 50.95 = 105$)

Effect of water table on effective stress

- Rising of WT reduces σ' (u increases, σ increases by smaller amount), leading to loss of strength and bearing capacity.
- Lowering of WT increases σ' and causes settlement of the ground (e.g. Mexico City, Bangkok subsidence due to pumping).

- Capillary rise above WT: u is negative, so $\sigma' = \sigma + \gamma_w h$ (increase) - apparent cohesion.
- A submerged soil: $\sigma' = z \gamma'$ (z = depth below the soil surface), independent of the water depth above the soil surface.

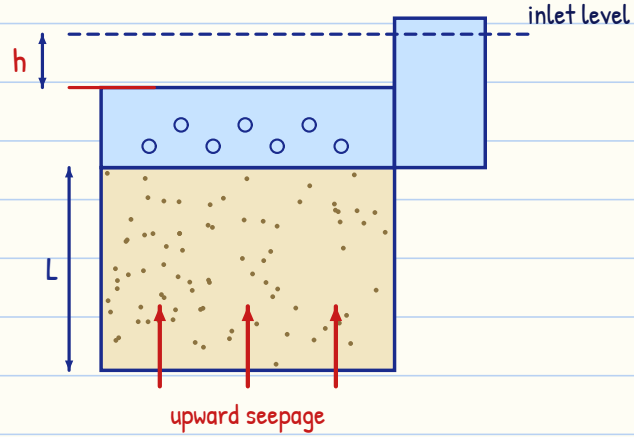
Pore pressure in a saturated soil under a static water table is hydrostatic. Water level above the ground changes both σ and u by the same amount, so σ' does not change.

8. Seepage and Seepage Pressure

When water flows through soil, it exerts a drag on the soil grains in the direction of flow, called the seepage force.

- Seepage pressure $p_s = h_L \gamma_w$ (pressure lost due to friction over the flow length) and seepage force per unit volume $j = i \gamma_w$.
- Downward seepage: σ' increases: $\sigma' = z \gamma' + i z \gamma_w$. (Helps stability.)
- Upward seepage: σ' decreases: $\sigma' = z \gamma' - i z \gamma_w = z (\gamma' - i \gamma_w)$. (Reduces stability.)

Quick Sand Condition



At the bottom of the sand layer:
total stress $\sigma = L \gamma_{sat} + z_w \gamma_w$
pore pressure $u = (L + h + z_w) \gamma_w$
$\sigma' = L \gamma' - h \gamma_w$

Quick condition: $\sigma' = 0$
$L \gamma' = h \gamma_w$
$i_c = h / L = \gamma' / \gamma_w$

As the upward hydraulic gradient increases, σ' becomes smaller. When the effective stress becomes zero, the soil grains lose contact with each other, the soil has no shear strength and behaves like a liquid - boiling of sand. This is quick sand condition (also called boiling, piping or liquefaction due to seepage).

$$\sigma' = 0 \Rightarrow i_c = \gamma' / \gamma_w = (G - 1) / (1 + e)$$

- i_c = critical hydraulic gradient; for most sands G = 2.65 and e = 0.6 - 0.8, i_c is about 1.0.
- Factor of safety against piping $F = i_c / i_{exit}$. In design F = 4 - 5 for the exit gradient (or $i_{exit} <= 0.2 - 0.25$).

- Quick sand is not a type of sand - it is a condition that can occur in any cohesionless soil (fine sand, silt) under upward flow.
- It happens: in excavations below water table, at downstream toe of dams and weirs, in river banks and cofferdams.

Remedies

- Reduce the exit gradient: longer path through cut-off walls (sheet piles), upstream impervious blanket.
- Provide filters / weighted inverted filter at the exit point, relief wells, and dewatering (wellpoints).

Example 4

A sand layer: G = 2.67, e = 0.65.

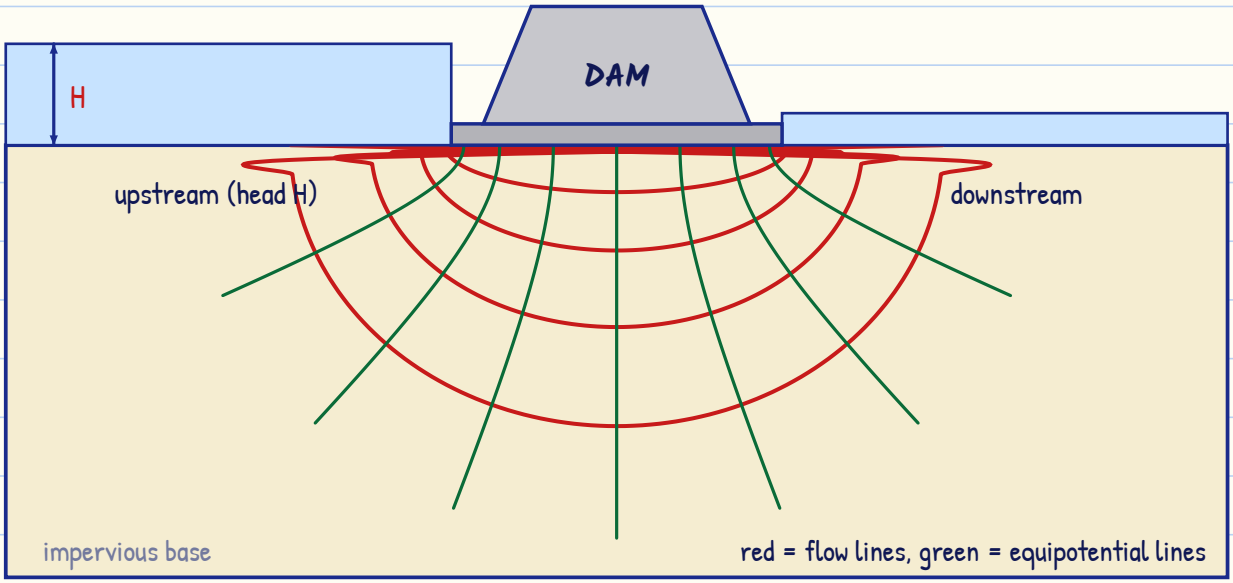
- $i_c = (2.67 - 1) / (1 + 0.65) = 1.67 / 1.65 = 1.01$.
- If a 2 m thick layer of this sand has an upward head of 1.5 m, i = 1.5/2 = 0.75, F = 1.01/0.75 = 1.35 (not safe; < 4).
- Head that would cause boiling: $h = i_c L = 1.01 \times 2 = 2.02$ m.

9. Flow Nets

Steady two-dimensional flow through isotropic soil is governed by Laplace's equation:

$$\partial^2 h/\partial x^2 + \partial^2 h/\partial z^2 = 0 \quad (\text{assuming } k_x = k_z)$$

- A flow net is the graphical solution to Laplace's equation: a net of flow lines and equipotential lines, intersecting at 90 degrees, forming curvilinear squares.
- Flow line (stream line): path followed by a water particle. Equipotential line: line joining points of equal total head.



Characteristics (properties) of a flow net

- Flow lines and equipotential lines meet at right angles (orthogonal).
- The fields (elements) formed are curvilinear squares - the average width equals the average length, so a circle can be inscribed.
- The quantity of seepage is the same in each flow channel; the head drop is the same between successive equipotentials.
- Flow lines do not cross each other; neither do equipotential lines.
- Smaller the size of a square, higher the hydraulic gradient and velocity there (e.g. near the toe).
- Impervious boundaries (base of the soil, sheet pile surface) are flow lines. Upstream and downstream soil surfaces under water are equipotential lines.

Seepage discharge

$$q = k\,H\,(N_f / N_d) \qquad \text{per unit width}$$

- H = total head loss; N_f = number of flow channels; N_d = number of equipotential drops (equipotential intervals). The ratio N_f/N_d is called the shape factor.
- Derivation: for one square field, $\Delta q = k\,(\Delta h / l) \times b = k\,\Delta h$ (since b = l); $\Delta h = H/N_d$; $q = N_f\,\Delta q$.

Pore pressure, uplift and exit gradient

- The head at the n-th equipotential: $h_n = H - n\,(H / N_d)$. Pressure head = total head - elevation head, so $u = \gamma_w\,(h_n - z)$.
- Exit gradient $i_e = (H / N_d) / l_e$, where l_e is the length of the last field at the exit. Safety against piping $F = i_c/i_e$.

- Uplift pressure on the base of a dam is obtained from the pore pressures read off the flow net.

Example 5

Flow net below a concrete dam: H = 8 m, N_f = 4, N_d = 12, k = 3 x 10^{-6} m/s.

- q = 3 x 10^{-6} x 8 x (4/12) = 8 x 10^{-6} m^3/s per metre length (= 0.69 m^3/day/m).
- Head drop per field: 8/12 = 0.667 m.
- If the last field at the toe is 1.5 m long, i_e = 0.667/1.5 = 0.44, with i_c = 1.0, F = 2.25 (not enough, add cut-off).

Construction of a flow net - steps

- Draw the structure and soil boundaries to a convenient scale. Identify boundary conditions.
- Sketch 3 - 4 flow lines, then draw equipotentials so as to get squares; adjust by trial and error until all fields are squares.
- Anisotropic soils: transform the section: $x' = x \sqrt{(k_z / k_x)}$, draw the net in the transformed section and use $k_e = \sqrt{(k_x k_z)}$ in $q = k_e H N_f/N_d$.
- Earth dams: the top flow line (phreatic line) is a basic parabola with focus at the toe of the dam (Casagrande); it is a flow line with u = 0.

Uses of flow nets

- Estimate seepage loss (quantity of seepage) under dams and through embankments.
- Determine pore water pressure at any point, hence effective stress and uplift on the base of structures.
- Find the exit gradient and check the safety against piping.
- Decide the depth of cut-off walls and the length of impervious floors.
- Check stability of slopes and design filters / drainage of earth dams.

Other methods of solution: analytical (Laplace), numerical (finite difference / finite element), electrical analogy and Casagrande's graphical method (the trial-and-error sketching method).

10. Quick Revision

Item	Formula
Capillary rise	$h_c = 4\,T\cos\alpha / (\gamma_w\, d) = C / (e\, D_{10})$
Darcy's law	$v = k\,i,\ q = k\,i\,A$
Seepage velocity	$v_s = v\,(1 + e) / e$
Constant head	$k = Q\,L / (A\,h\,t)$
Falling head	$k = 2.303\,(a\,L / A\,t)\log(h_1/h_2)$
Layered, horizontal	$k_h = \Sigma\, k\,z / \Sigma\, z$
Layered, vertical	$k_v = \Sigma\, z / \Sigma\,(z/k)$
Effective stress	$\sigma' = \sigma - u$
Critical gradient	$i_c = (G - 1)/(1 + e) = \gamma'/\gamma_w$
Flow net discharge	$q = k\,H\,N_f / N_d$

Important questions (exam)

- Explain capillary rise in soils and the effect of capillary tension on effective stress.
- State Darcy's law and its limitations. Differentiate discharge and seepage velocity.
- What are the factors affecting permeability of soils?
- Describe the constant head and falling head permeability tests, with the formulae.
- Derive the equivalent permeability for flow parallel and perpendicular to layers.
- Explain Terzaghi's effective stress principle. What is quick sand condition? Derive critical hydraulic gradient.
- What are flow nets? State the characteristics and uses; derive $q = k\,H\,N_f/N_d$.

GEOTECHNICAL ENGINEERING

UNIT - III : STRESS DISTRIBUTION IN SOILS & COMPACTION

Course Code: 23A01503T

Syllabus: STRESS DISTRIBUTION IN SOILS: Boussinesq's and Westergaard's theories for point loads and areas of different shapes - Newmark's influence chart. COMPACTION: Mechanism of compaction - Factors affecting - Effects of compaction on soil properties - Field compaction equipment - Compaction control.

PART A : STRESS DISTRIBUTION IN SOILS

1. Introduction

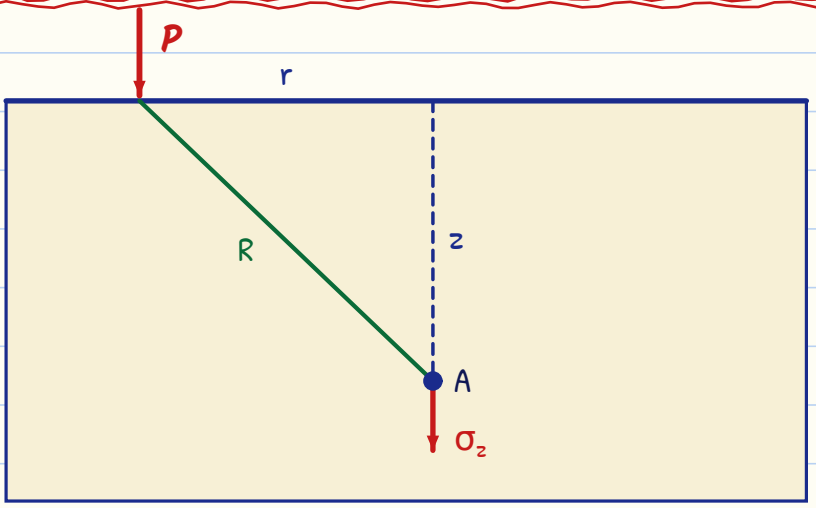
Loads from structures are transferred to the soil through foundations. The stress increase $\Delta\sigma$ at different depths below the foundation is needed to compute settlement and to check the shear failure of the soil.

- Soil is not a perfectly elastic material, but the theory of elasticity is used because it gives reasonably good results in practice (stresses are not much affected by the modulus).
- Assumptions of the elastic theories: the soil mass is homogeneous, isotropic, linearly elastic and extends to infinite depth (semi-infinite half space) with a horizontal surface; weight of the soil is ignored (only the increase in stress due to the load is computed); the loaded area is on the surface and the contact stress is known.
- Boussinesq (1885): homogeneous, isotropic, elastic half space. Westergaard (1938): elastic material reinforced by thin, rigid horizontal sheets which prevent lateral strain (stratified soils).

Types of loads

- Point (concentrated) load P. Line load q per unit length.
- Strip load (long footing, wall footing). Circular and rectangular uniformly loaded areas.
- Embankment (triangular) load, etc.

2. Boussinesq's Theory for a Point Load



Boussinesq's equation

$\sigma_z = 3P\, z^3 / (2\pi R^5)$

$= I_B\, P / z^2$

$I_B = (3/2\pi)\,[1/(1+(r/z)^2)]^{5/2}$

$R = \text{sqrt}\,(r^2 + z^2)$

max. at $r = 0$ ($I_B = 0.4775$)

For a concentrated load P acting on the surface of a semi-infinite elastic mass, the vertical stress at a point A at a depth z and radial distance r is

$$\sigma_z = (3P / 2\pi)\, z^3 / R^5 = I_B\, P / z^2$$

- $R^2 = r^2 + z^2$. The influence factor $I_B = (3 / 2\pi)\,[\,1 / (1 + (r/z)^2)\,]^{5/2}$ depends only on r/z.
- The stress is independent of the elastic constants (E and ν) in Boussinesq's equation for σ_z.
- Directly under the load ($r = 0$): $\sigma_z = 0.4775\, P / z^2$. The stress decreases rapidly with the depth and with the radial distance.

r/z	0	0.25	0.5	1.0	1.5	2.0	3.0
I_B (Boussinesq)	0.4775	0.4103	0.2733	0.0844	0.0251	0.0085	0.0015

Example 1

A point load of 500 kN acts on the surface. Find σ_z at 3 m depth and 2 m horizontal distance from the load.

- $r/z = 2/3 = 0.667$; $I_B = 0.4775 \times [1/(1 + 0.444)]^{2.5} = 0.4775 \times 0.399 = 0.190$
- $\sigma_z = 0.190 \times 500 / 3^2 = 10.6$ kPa

Other results of Boussinesq

- Line load q per unit length: $\sigma_z = 2\, q\, z^3 / [\pi\, (x^2 + z^2)^2]$.
- Uniformly loaded circular area of radius R (below the centre):

$$\sigma_z = q\,[\,1 - 1 / (1 + (R/z)^2)^{3/2}\,]$$

- Strip load of intensity q and width B: $\sigma_z = (q/\pi)\,[\,\alpha + \sin\alpha \cos(\alpha + 2\delta)\,]$, where α is the angle subtended at the point by the strip and δ is the angle between the vertical and

the line to the nearer edge.

- Rectangular area: $\sigma_z = q\, I_r$ below a corner.

Example 2 (circular footing)

q = 150 kPa, R = 2 m, z = 3 m. $(R/z)^2 = 0.444$; $1/(1.444)^{1.5} = 0.576$; $\sigma_z = 150\,(1 - 0.576) =$ 63.6 kPa.

3. Westergaard's Theory

Westergaard (1938) considered soils such as stratified deposits (varved clays, sand layers with clay seams). The soil is assumed to be an elastic medium in which thin, rigid sheets (no deformation) prevent lateral strain. For Poisson's ratio $\nu = 0$:

$$\sigma_z = (P / z^2) \times (1/\pi) / [\,1 + 2\,(r/z)^2\,]^{3/2} = I_W\, P / z^2$$

- Directly under the load: $I_W = 1/\pi = 0.3183$ (compared with $I_B = 0.4775$).

r/z	0	0.5	1.0	1.5	2.0	3.0
I_W (Westergaard)	0.3183	0.1733	0.0613	0.0247	0.0118	0.0038

Boussinesq vs Westergaard

Aspect	Boussinesq	Westergaard
Soil assumed	homogeneous, isotropic, elastic	stratified, thin rigid sheets (no lateral strain)
Depends on ν	no	yes (ν = 0 usually)
Stress below load	larger ($0.4775\ P/z^2$)	smaller ($0.3183\ P/z^2$)
Suitable for	homogeneous clay / sand	stratified, varved deposits

Boussinesq gives higher stress near the axis and is more conservative (safe). It is mostly used in practice. The Westergaard values are about 2/3 of the Boussinesq values under the load.

4. Stresses Under Areas of Different Shapes

Rectangular area (Fadum / Newmark)

Below the corner of a uniformly loaded rectangle B x L (B = width, L = length) at a depth z:

$$\sigma_z = q\, I_r \qquad I_r = f(m, n),\ m = B / z,\ n = L / z$$

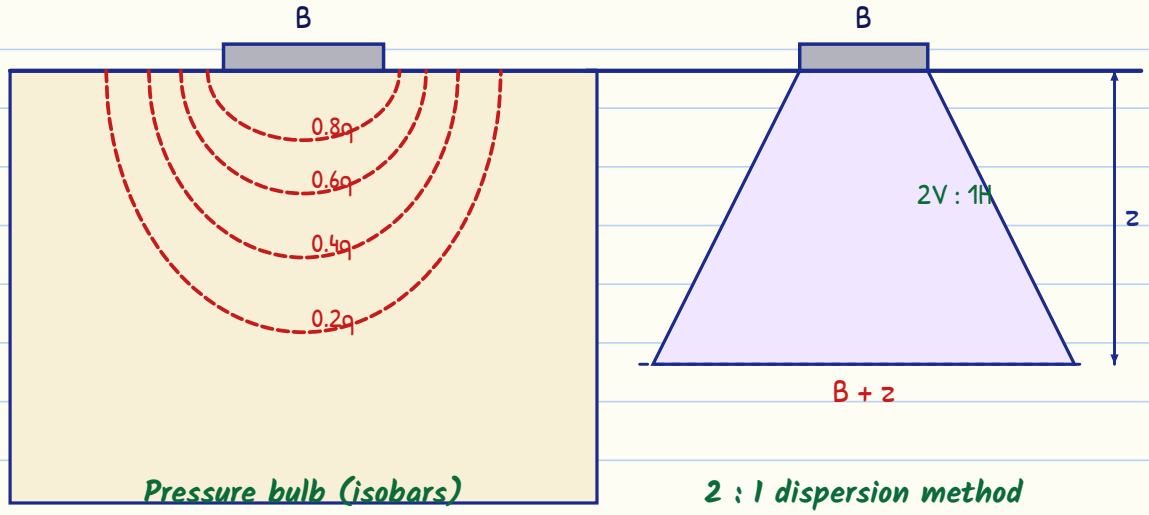
- I_r is given by charts / tables (Fadum chart); $I_r = (1/4\pi)\,[\,(2mn\,\mathrm{sqrt}(m^2+n^2+1)/(m^2+n^2+m^2n^2+1))\times((m^2+n^2+2)/(m^2+n^2+1)) + \tan^{-1}(2mn\,\mathrm{sqrt}(m^2+n^2+1)/(m^2+n^2+1-m^2n^2))\,]$
- For a point not at the corner use the superposition method: divide the area into rectangles with a common corner above the point and add (or subtract) the effects.
- Below the centre: divide into 4 equal rectangles, so $\sigma_z = 4\,q\,I_r(B/2, L/2)$.

Example 3

A 4 m x 2 m rectangular footing carries q = 200 kPa. Find σ_z at 2 m below (i) a corner, (ii) the centre.

- (i) $m = B/z = 2/2 = 1$, $n = L/z = 4/2 = 2$, $I_r = 0.1999$. $\sigma_z = 200 \times 0.1999 = 40$ kPa.
- (ii) Four rectangles of 2 m x 1 m: $m = 1/2 = 0.5$, $n = 2/2 = 1.0$, $I_r = 0.1202$. $\sigma_z = 4 \times 200 \times 0.1202 = 96.2$ kPa.

Approximate (2 : 1 distribution) method



The load is assumed to spread at a slope of 2 vertical : 1 horizontal so that the width increases by z at depth z.

Strip: $\sigma_z = q\,B / (B + z)$ Rectangle: $\sigma_z = q\,B\,L / [\,(B + z)(L + z)\,]$

- Simple, but gives the average stress; it is fairly accurate between $z = B$ and $4B$.

Example 4

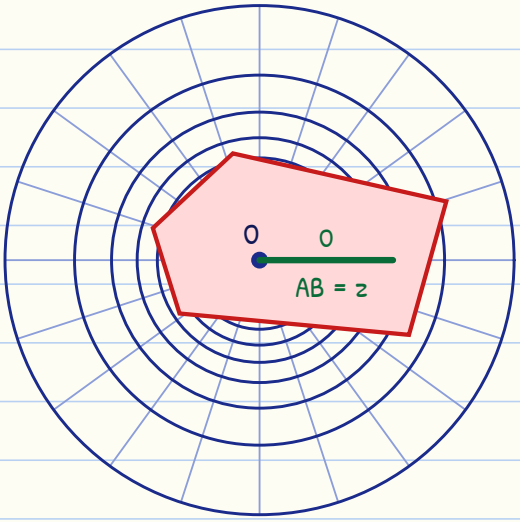
Square footing 2 m x 2 m, q = 200 kPa, find the stress at z = 3 m: $\sigma_z = 200 \times 2 \times 2 / (5 \times 5) = 32$ kPa.

Pressure bulb (isobar diagram)

- An isobar is a contour joining points of equal vertical stress below a loaded area. The pressure bulb is usually taken as the isobar of 0.2 q (20% of contact pressure) or 0.1 q.

- Significant depth (depth of influence): about 1.5 B for strip and 2 B for square footings (the bulb of 0.2 q).
- Closely spaced footings have overlapping bulbs, so the stresses add up; this is why a large foundation affects deeper soil than a small one, and settlement depends on the size of foundation.

5. Newmark's Influence Chart



Newmark chart:

9 concentric circles + 20 radial lines
(10 circles with the centre)
radii / z = 0.270, 0.401, 0.518, 0.637, 0.766, 0.918, 1.110, 1.387, 1.908

200 elements, each influence value
I = 1/200 = 0.005

AB = scale line = depth z
Footing (red) drawn to the scale AB = z with the point O below which σ_z is wanted placed at the chart centre.

Newmark (1942) developed a chart based on Boussinesq's equation for a circular loaded area, to get the vertical stress below any irregular shaped loaded area at any point.

- For a circular area of radius R carrying q, the stress ratio $\sigma_z/q = 1 - [1/(1 + (R/z)^2)]^{3/2}$. Choose σ_z/q = 0.0, 0.1, 0.2 ... 0.9, 1.0 and solve for R/z; this gives the radii of the concentric circles. (For σ_z/q = 1, R/z = infinity.)
- Then the circles are divided by 20 equally spaced radial lines, so the 10 rings x 20 = 200 elements, each having the same influence value I = 0.1 / 20 = 0.005.

Procedure

- Fix the depth z at which the stress is required and draw the plan of the loaded area to a scale such that the length of the scale line AB on the chart equals z.
- Place the plan so that the point O (vertically above which stress is required) coincides with the centre of the chart.

- Count the number of elements N covered by the plan (count partial elements by estimate).

$$\sigma_z = I \times N \times q = 0.005\, N\, q$$

- For the stress at another depth, change the scale of the plan (AB = new z). For another point, shift the plan.

Example 5

For a footing plan placed on the chart (AB = z) the number of elements covered = 62 and q = 180 kPa: σ_z = 0.005 x 62 x 180 = 55.8 kPa.

Advantages

- Can be used for loaded areas of any shape, including multiple or irregular footings, and for any point inside or outside.
- Simple to use, but a new plan has to be drawn for each depth.

PART B : COMPACTION

6. Introduction and Mechanism of Compaction

Compaction is the process of densification of soil by the application of mechanical energy (rolling, tamping, vibration), which reduces the air voids at constant water content.

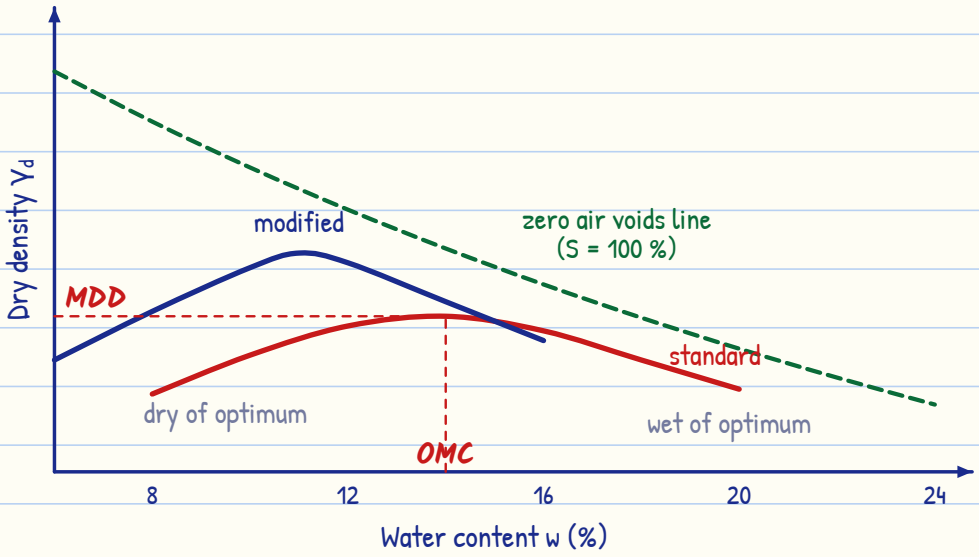
- Aims: increase shear strength, reduce compressibility (settlement), reduce permeability, reduce swelling and shrinkage and resist frost / liquefaction.
- Applications: embankments, earth dams, road subgrade and base, backfills, airfield pavements, and ground improvement.

Compaction	Consolidation
Expulsion of air by mechanical energy	Expulsion of water by static load
Short term, quick (during construction)	Long term, slow (years)
Water content constant	Water content decreases
Applies to partly saturated soil	Applies to saturated soil

Mechanism (R.R. Proctor, 1933)

- Dry soil: particles have friction and no lubrication, so air voids remain and the dry density is low.

- With a small amount of water added, water acts as a lubricant (water films) and the particles slide into closer packing, so dry density increases.
- At the optimum water content (OMC) the dry density is maximum (MDD). The air voids are at a minimum, about 5 - 10 %.
- Beyond the OMC, additional water fills the voids and keeps the grains apart; the air cannot be expelled and the pore pressure builds up, so the dry density decreases. The curve can never exceed the zero air voids line (S = 100 %).



Laboratory compaction tests (Proctor test)

Item	Standard (Light) Proctor	Modified (Heavy) Proctor
IS code	IS 2720 Part 7	IS 2720 Part 8
Mould	1000 cc (100 mm dia, 127.3 mm high)	1000 cc (same)
Rammer mass	2.6 kg	4.9 kg
Height of fall	310 mm	450 mm
Layers x blows	3 x 25	5 x 25
Energy	about 592 kJ/m^3	about 2700 kJ/m^3

- Soil passing the 19 mm / 4.75 mm sieve is mixed at 4 - 5 different water contents; each sample is compacted in the mould; bulk density γ and the water content w are found and the dry density is computed:

$$\gamma_d = \gamma / (1 + w)$$

- Plot γ_d against w to get the compaction curve. The peak gives the maximum dry density (MDD) and the optimum moisture content (OMC).
- Zero air voids line / saturation line: $\gamma_d = G \gamma_w / (1 + w G)$. For a given percentage of air voids $\gamma_d = (1 - n_a) G \gamma_w / (1 + w G)$.

Example 6

Standard Proctor test data (G = 2.70):

w (%)	10	12	14	16	18
γ bulk (g/cc)	1.89	2.01	2.08	2.06	2.00
$\gamma_d = \gamma/(1+w)$	1.718	1.795	1.825	1.776	1.695

- OMC = 14 %, MDD = 1.825 g/cc. ZAV at 14 %: 2.70/(1 + 0.14 x 2.70) = 1.959 g/cc (above the curve, as it should be).
- At OMC: e = 2.70/1.825 - 1 = 0.479; S = wG/e = 0.14 x 2.70/0.479 = 0.789 = 78.9 % (so about 21 % of the voids hold air).

7. Factors Affecting Compaction

- Water content: the most important factor - as explained above, MDD occurs only at OMC.
- Compaction effort (energy): greater effort gives a higher MDD and lower OMC (curve shifts upwards and to the left, as in the figure, modified Proctor vs standard). It is more effective on the dry side. Beyond a limit, increase in effort gives little gain.

  E = (blows x layers x rammer weight x fall height) / V

- Type of soil (gradation, grain size): well-graded coarse soils reach high MDD (2.0 - 2.3 g/cc) at low OMC (6 - 10 %); uniform sands give a flat curve; clays give lower MDD (1.4 - 1.8 g/cc) at high OMC (15 - 30 %) with a sharply peaked curve for low plasticity, a flat curve for high plastic clay.
- Method of compaction: impact (Proctor), kneading (sheepsfoot), static pressure, vibration - for sands vibration is best; for clay kneading and impact. Laboratory results differ from field results.
- Admixtures: lime, cement, bitumen and salts alter the OMC and MDD; coarse particle content (oversize) affects the results (correction for the oversize particles).
- Lift thickness and number of passes: thick layers are not compacted at the bottom.
- Temperature, initial water content, and soil structure have minor effects.

Soil type	Typical OMC (%)	Typical MDD (g/cc)
Well-graded sand / gravel	6 - 10	2.0 - 2.3
Silty sand	9 - 14	1.85 - 2.1
Silt / sandy clay	12 - 20	1.65 - 1.95
Clay (CL / CH)	20 - 30	1.4 - 1.7

Clean sand and gravel: the dry density is highest when completely dry or fully saturated (bulking in between), so compaction is controlled by relative density and flooding / vibration, rather than by OMC.

8. Effects of Compaction on Soil Properties

Property	Dry side of OMC	Wet side of OMC
Structure	flocculated (random, higher strength)	dispersed (oriented particles)
Permeability	higher	lowest at about OMC, further lower
Compressibility (low pressure)	lower (stiffer)	higher
Compressibility (high pressure)	higher (structure collapses)	lower
Shear strength (as compacted)	higher (suction)	lower
Swelling	higher (large swell on saturation)	lower
Shrinkage	lower	higher
Pore water pressure	low (negative)	high (positive)

- Increase in dry density leads to higher strength (CBR increases), lower permeability and lower settlement.
- Compacting dry of optimum gives a high strength soil, but if it later gets wet it may swell or collapse. For earth dam core and canal lining use wet of optimum (low permeability, flexible, tolerates differential settlement without cracking). For road subgrade use at or slightly dry of optimum (high CBR).
- Pore pressures and anisotropy develop during compaction in clays; the horizontal permeability is greater than the vertical (layered effect).
- Compaction increases the density and reduces the settlement of the sand and also reduces the risk of liquefaction.

9. Field Compaction Equipment

Equipment	Suitable soils / uses
Smooth-wheeled (steel drum) roller, static 8 - 10 t	sand, gravel, crushed stone, sub-base, bituminous layers (not for clay at depth)
Sheepsfoot (tamping foot) roller	cohesive soils (clays, silty clays), kneading action; foot pressure 1.5 - 7 MPa
Pneumatic tyred roller (heavy 10 - 50 t)	most soils, both sand and clay; also asphalt; adjustable tyre pressure
Vibratory roller (drum with vibrator)	granular, cohesionless soil (sand, gravel, rockfill); good for thick lifts
Vibrating plate / rammer / frog rammer	small and confined areas, trenches, near walls
Dynamic compaction (heavy tamping) 10 - 40 t weight dropped 10 - 40 m	deep compaction of loose granular fill, up to 10 m depth
Vibroflotation (vibro compaction)	clean sand deep layers, ground improvement

- Steel-wheel rollers give a smooth surface but the contact pressure is low at the bottom of the lifts.
- Sheepsfoot rollers compact from the bottom to the top, as the feet gradually "walk out" of the layer.
- Loose layer thickness: about 150 - 300 mm for rollers, 100 - 150 mm for the plate vibrators; the number of passes is generally 6 - 12 (found by a trial test strip).
- Choose the equipment from soil type, size of the job and thickness of the lift. Moisture content must be near OMC (add water by sprinkling, or dry the soil by aeration / blending).

10. Compaction Control

The aim is to ensure that the compacted fill meets the specified density and water content.

The control is done by two methods:

(a) Method specification (end product not tested)

- The type and weight of the roller, the thickness of the lift, the number of passes and the water content are specified, from a trial section (test strip).

(b) End product specification (more common)

- The relative compaction (degree of compaction) of the field-compacted soil is specified:

$$R.C. = \left(\gamma_{d,\ field} / \gamma_{d\ max,\ lab} \right) \times 100 >= 95\ \%\ \text{(often 95 - 100 \%)}$$

- Typical: embankments 95 %, earth dam core 98 - 100 % of standard Proctor MDD, road subgrade 97 % of modified Proctor.

- Water content to be maintained within OMC +/- 2 % (dry or wet side as chosen).
- For cohesionless soils (sands): control by relative density D_r >= 70 %.

Field density methods for control

- Core cutter method (clay), sand replacement method (gravel and sand), rubber balloon, nuclear density gauge (quick and widely used) and proctor needle (quick check of water content / strength).
- Frequency: one test per 1000 m^2 per layer (as in the specification) and at least 3 tests per lift.
- Check for the air void percentage (should be less than 5 - 10 %) and the saturation line: points above the ZAV line are wrong.

Example 7

Field density tests give a bulk density of 1.98 g/cc at a water content of 13 %. MDD = 1.85 g/cc.

- γ_d = 1.98/1.13 = 1.752 g/cc; R.C. = 1.752/1.85 = 0.947 = 94.7 % (below the 95 % requirement, so recompact the layer).

Problems in the field

- Over-compaction (too many passes, or too wet soil) causes "rubbering / pumping" and shearing of the layer.
- Soft spots on wet clay: dry the soil by aeration or add lime.
- Laminations between the layers: scarify the surface of the previous layer before laying the next lift.

II. Quick Revision

Item	Formula
Boussinesq point load	$\sigma_z = I_B P / z^2$, $I_B = (3/2\pi)[1/(1+(r/z)^2)]^{5/2}$
Westergaard point load	$\sigma_z = I_W P / z^2$, $I_W = (1/\pi)/[1+2(r/z)^2]^{3/2}$
Circular area (centre)	$\sigma_z = q\,[1 - (1/(1+(R/z)^2))^{3/2}]$
Rectangle (corner)	$\sigma_z = q\, I_r(m, n)$
2:1 method	$\sigma_z = q\, B\, L / [(B+z)(L+z)]$
Newmark chart	$\sigma_z = 0.005\, N\, q$
Dry density	$\gamma_d = \gamma/(1+w)$

ZAV line	$\gamma_d = G \gamma_w / (1 + w G)$
Relative compaction	R.C. = $\gamma_{d\ field} / \gamma_{d\ max}$

Important questions (exam)

- Derive / state Boussinesq's equation for vertical stress below a point load. How does Westergaard's analysis differ?
- Explain how stress beneath a rectangular area is found. What is the 2:1 method?
- What is Newmark's influence chart? Explain its construction and use.
- Explain the mechanism of compaction. Explain the standard Proctor test and the compaction curve.
- Discuss the factors that affect compaction and the effects of compaction on soil properties.
- Describe the field compaction equipment and the methods of compaction control.

GEOTECHNICAL ENGINEERING

UNIT - IV : CONSOLIDATION OF SOILS

Course Code: 23A01503T

Syllabus: CONSOLIDATION: Types of compressibility - Primary consolidation and secondary consolidation - Stress history of clay; e-p and e-log p curves - Normally consolidated soil, over consolidated soil and under consolidated soil - Pre-consolidation pressure and its determination - Terzaghi's 1-D consolidation theory - Coefficient of consolidation: square root time and logarithm of time fitting methods.

1. Introduction

Under a load the soil decreases in volume - this is called compression. When a structure is built on a clay layer the clay takes years to settle, whereas in sand the settlement is complete as soon as the load is applied.

- Compressibility = the property by which the volume of the soil decreases under the action of compressive load.
- Consolidation = the gradual reduction in volume of a fully saturated soil of low permeability due to the drainage of pore water under a sustained static load. It is a time-dependent process.
- Volume decrease in a soil is due to: (i) compression of solid grains and water (very small, ignored), (ii) compression of air in voids (partially saturated soils), (iii) expulsion of water (and air) from the voids - the main reason.
- Consolidation is important because it governs the settlement of structures on clay (leaning tower of Pisa, Mexico City palace, Bangkok).

Types of settlement (compressibility)

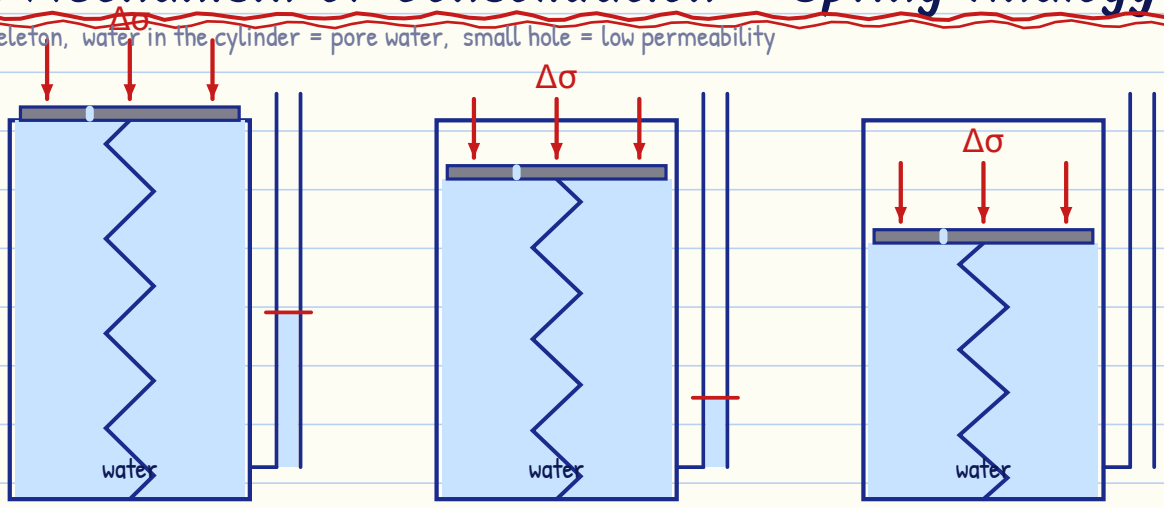
Type	Cause	Time
Immediate (initial) settlement S_i	elastic distortion of soil without change in water content (shape change)	immediately on loading
Primary consolidation S_c	expulsion of water from the voids, transfer of load from water to the soil skeleton	days to years
Secondary consolidation S_s	plastic readjustment of the particles at constant effective stress, after the excess pore pressure vanishes	continues for years

$$S_{total} = S_i + S_c + S_s$$

- For coarse-grained soils, consolidation is practically instant, so the settlement is mainly immediate. For saturated clays, S_c is the main component, and organic soil and peat have large S_s.

2. Mechanism of Consolidation - Spring Analogy

spring = soil skeleton, water in the cylinder = pore water, small hole = low permeability



At t = 0
u = Δσ, σ' = 0

0 < t < ∞
u + σ' = Δσ

t = ∞
u = 0, σ' = Δσ

Terzaghi explained the process by a piston-spring-cylinder model: a cylinder filled with water, a piston with a small hole (low permeability of clay) and a spring (soil skeleton).

- At t = 0: load Δσ is applied; water cannot escape instantly, is incompressible, so water takes the whole load: excess pore pressure u = Δσ, effective stress σ' = 0. The spring is not compressed.
- 0 < t < infinity: water escapes through the hole, the piston moves down, the spring compresses and carries a part of the load: σ' increases and u decreases, such that Δσ = σ' + u.
- At t = infinity: excess pore pressure fully dissipated, u = 0 and σ' = Δσ; the spring carries the whole load. Primary consolidation is complete.

Consolidation = dissipation of excess pore water pressure with a corresponding increase in effective stress. Rate of consolidation depends on the permeability and the drainage path length, not on the magnitude of the load.

3. Laboratory Consolidation Test (Oedometer)

IS 2720 (Part 15). Gives the e - log p curve, the compressibility parameters (C_c, a_v, m_v) and the coefficient of consolidation c_v.

- Apparatus: consolidometer (fixed ring or floating ring cell), loading frame (lever system), porous stones (top and bottom, for double drainage), dial gauge (0.002 mm least count), water bath.
- Specimen: undisturbed soil, in a ring of 60 mm dia (or 70 mm) and 20 mm height (ratio dia/height at least 3, to minimise side friction).
- Procedure: the specimen is kept in the cell with the stones, saturated by flooding. Load increments are applied with $\Delta p/p = 1$ (0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 3.2 kg/cm^2), each load being kept for 24 hours (until the primary consolidation is complete). The dial readings are taken at 0.25, 1, 2.25, 4, 6.25, 9, 12.25, 16, 20.25, 25, 30.25 ... min (square numbers, for the root time method) and then at 24 h.
- After the maximum load the load is reduced in steps (unloading, rebound) and then reloaded to find the preconsolidation pressure (optional).

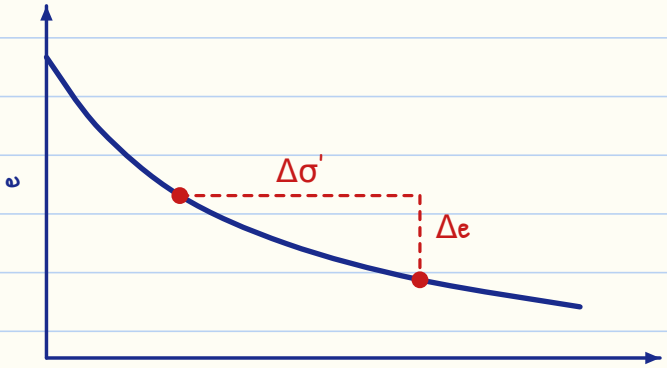
Calculation of void ratio

- Height of solids: $H_s = W_s / (A\ G\ \gamma_w)$. Initial void ratio $e_0 = (H_0 - H_s) / H_s$ (or $e_0 = w\ G$ for a saturated soil).

$$e = e_0 - \Delta H\ (1 + e_0) / H_0$$

- At the end of every load increment, find the final thickness H and the void ratio e: $e = (H - H_s) / H_s$. Plot e vs σ' (arithmetic or log scale).

4. e - p and e - log p Curves and Compressibility Parar



Coefficient of compressibility
$a_v = - \Delta e / \Delta \sigma'$ (m^2/kN)

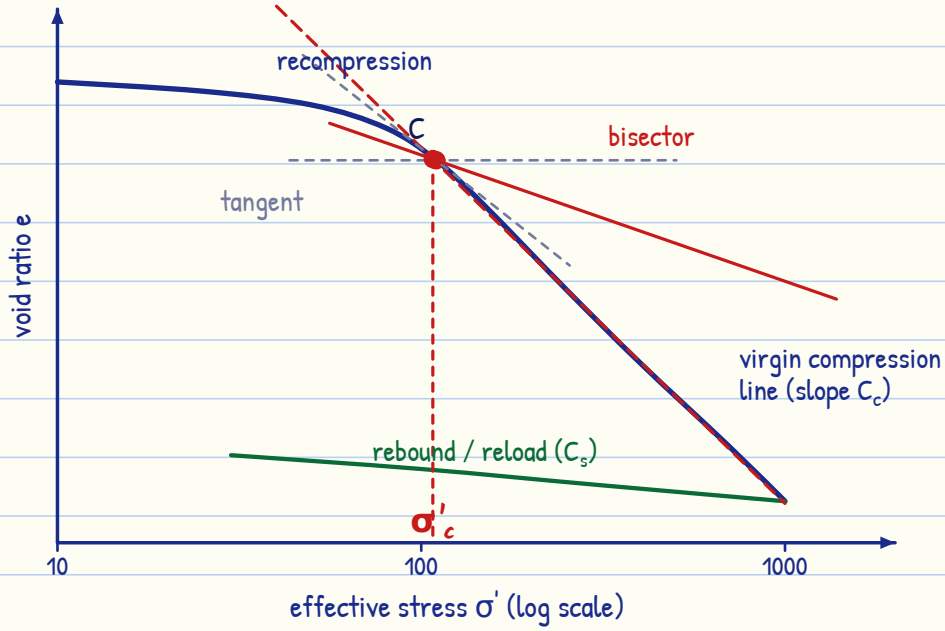
Coeff. of volume compressibility
$m_v = a_v / (1 + e_0)$

Compression index (e - log σ' plot)
$C_c = - \Delta e / \Delta \log \sigma'$

Swelling index C_s = 1/5 to 1/10 of C_c

- Coefficient of compressibility $a_v = - \Delta e / \Delta \sigma'$: slope of the e - p curve, decreases with increasing pressure; typical NC clay: 0.2 - 1 x 10^{-3} m^2/kN. (Note: negative sign because e decreases.)
- Coefficient of volume compressibility $m_v = a_v / (1 + e_0) = \Delta H / (H_0 \Delta \sigma') =$ volume change per unit initial volume per unit increase in pressure.

$$S_c = m_v \Delta \sigma' H_0 = [\, a_v / (1 + e_0) \,] \Delta \sigma' H_0 = [\, \Delta e / (1 + e_0) \,] H_0$$



- On the e - log σ' plot, the later part of the curve (beyond the preconsolidation pressure) is a straight line, the virgin compression line (VCL). Its slope is the compression index C_c.

$$C_c = - \Delta e / \Delta(\log_{10} \sigma') = (e_1 - e_2) / \log_{10} (\sigma'_2 / \sigma'_1)$$

- The unloading / reloading curve (recompression) has a flatter slope: the swelling (recompression) index C_s = 1/5 to 1/10 of C_c.

- Typical C_c: sand and gravel (very small), NC clays 0.2 – 0.5, soft clays 0.5 – 1.0, organic clay and peat > 1.0. High C_c means high compressibility.
- Empirical correlations: Terzaghi and Peck (undisturbed clay): $C_c = 0.009\,(w_L - 10)$; remoulded clay: $C_c = 0.007\,(w_L - 10)$. Also $C_c = 0.54\,(e_0 - 0.35)$ (Nishida) and $C_c = 0.0054\,(2.6\,w_n \ldots)$ approx.

Settlement of a normally consolidated clay layer

$$S_c = [\, C_c H / (1 + e_0) \,] \log_{10} [\, (\sigma'_0 + \Delta\sigma) / \sigma'_0 \,]$$

- H = thickness of the clay layer, σ'_0 = effective overburden pressure at the middle of the layer, $\Delta\sigma$ = increase in vertical stress at the middle of the layer.

Example 1

NC clay layer 4 m thick, $e_0 = 1.10$, $C_c = 0.30$, σ'_0 at the mid-depth = 80 kPa, $\Delta\sigma = 60$ kPa.

- $S_c = (0.30 \times 4 / 2.10) \times \log (140/80) = 0.571 \times 0.243 = 0.139$ m = 139 mm.

Example 2 (compression index)

A clay with $w_L = 52$ % (undisturbed): $C_c = 0.009\,(52 - 10) = 0.378$.

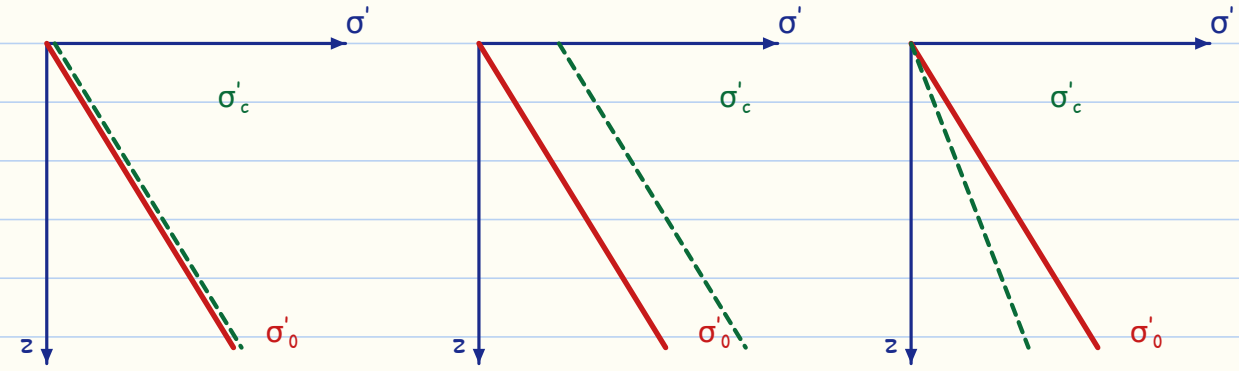
Example 3 (from void ratios)

Void ratio decreases from 0.92 to 0.80 when the pressure changes from 100 to 200 kPa. $a_v = 0.12/100 = 1.2 \times 10^{-3}$ m^2/kN; $C_c = 0.12 / \log 2 = 0.12/0.301 = 0.399$; $m_v = 1.2 \times 10^{-3} / 1.92 = 6.25 \times 10^{-4}$ m^2/kN.

5. Stress History of Clay

The stress history is the record of the maximum effective stress the soil has been subjected to in the past. The soil remembers the maximum past pressure.

- Preconsolidation pressure (σ'_c or p_c): the maximum effective stress experienced by the soil in its past.
- Over-consolidation ratio OCR = σ'_c / σ'_0, where σ'_0 = present effective overburden pressure.



Normally consolidated OCR = 1 | *Over-consolidated* OCR > 1 | *Under-consolidated* OCR < 1

(a) Normally consolidated (NC) soil

- Present effective overburden = the maximum past pressure: $\sigma'_0 = \sigma'_c$, OCR = 1. The soil has never been subjected to a pressure greater than the present one and is fully consolidated under the present overburden. Example: recently deposited alluvium and marine clay.

(b) Over-consolidated (OC) soil

- $\sigma'_0 < \sigma'_c$: OCR > 1. The soil was subjected in the past to a greater pressure than the present.
- Causes: erosion of overburden, melting of glaciers (ice load removed), lowering of the water table, desiccation (drying and shrinkage - crust formation), tectonic pressures, excavation, and secondary compression (ageing).
- Behaviour: stiffer, less compressible, higher strength, lower void ratio than an NC clay at the same depth. On loading the soil follows the flat recompression curve (C_s) until σ'_c and then the VCL.

(c) Under-consolidated (UC) soil

- The soil is still consolidating under the present overburden; excess pore pressure not yet dissipated, so $\sigma'_0 > \sigma'_c$ (OCR < 1). Example: recently placed fills and rapidly deposited sediments, e.g. delta deposits, dredged fill; highly compressible and weak.

State	OCR	σ'_0 compared with σ'_c	Remark
Normally consolidated	= 1	equal	on the VCL
Over-consolidated	> 1	$\sigma'_0 < \sigma'_c$	stiff, small settlement
Under-consolidated	< 1	$\sigma'_0 > \sigma'_c$	settling under own weight

Settlement of an over-consolidated clay

- If $\sigma'_0 + \Delta\sigma <= \sigma'_c$ (the final pressure remains below the preconsolidation pressure):

$$S_c = [\, C_s H / (1 + e_0) \,] \log [\, (\sigma'_0 + \Delta\sigma) / \sigma'_0 \,]$$

- If $\sigma'_0 < \sigma'_c < \sigma'_0 + \Delta\sigma$ (crosses the preconsolidation pressure):

$$S_c = [C_s H/(1+e_0)] \log (\sigma'_c/\sigma'_0) + [C_c H/(1+e_0)] \log [(\sigma'_0+\Delta\sigma)/\sigma'_c]$$

Example 4

OC clay 3 m thick, $e_0 = 0.95$, $C_c = 0.28$, $C_s = 0.05$, $\sigma'_0 = 60$ kPa, $\sigma'_c = 100$ kPa, $\Delta\sigma = 80$ kPa. Final pressure 140 kPa > σ'_c, so:

- $S_c = (0.05 \times 3/1.95) \log (100/60) + (0.28 \times 3/1.95) \log (140/100) = 0.0769 \times 0.2218 + 0.4308 \times 0.1461 = 0.01706 + 0.06294 = 0.080$ m = 80 mm. OCR = 100/60 = 1.67.

6. Preconsolidation Pressure and its Determination

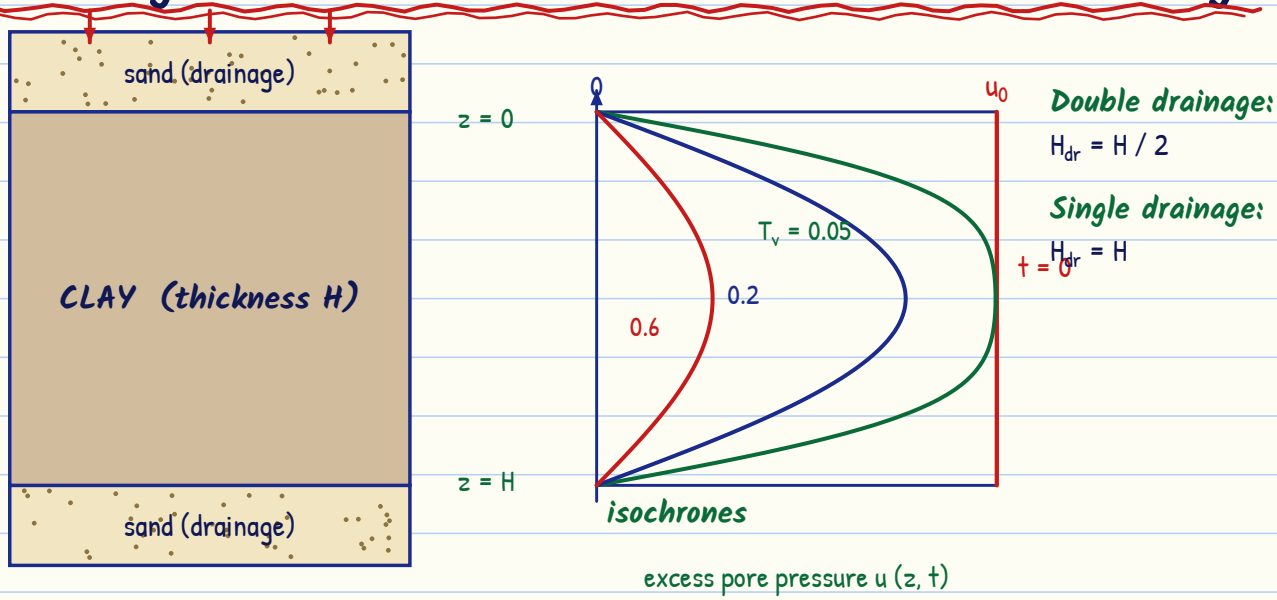
Casagrande's graphical method (1936) on the e - log σ' curve (see the figure above):

- Step 1: Pick the point C of maximum curvature (smallest radius) on the e - log p curve.
- Step 2: Draw the horizontal line through C and the tangent at C.
- Step 3: Draw the bisector of the angle between the horizontal line and the tangent.
- Step 4: Extend the straight line portion of the virgin compression curve backwards till it meets the bisector.
- Step 5: The abscissa of the point of intersection gives the preconsolidation pressure σ'_c.
- Other methods: Schmertmann's construction, Burmister, and the work-per-unit-volume (Becker) method.

Notes

- Sample disturbance rounds off the curve and so reduces the apparent σ'_c and flattens the VCL - the Schmertmann method corrects for it.
- OCR is found as σ'_c/σ'_0; the soil is NC if σ'_c is about equal to σ'_0.
- In design, if the structure pressure exceeds σ'_c the settlement will be large (use the VCL slope C_c).

7. Terzaghi's One-Dimensional Consolidation Theory



Terzaghi (1925) gave the theory for the rate of consolidation of a saturated clay layer under a sudden load, with drainage and compression in the vertical direction only.

Assumptions

- The soil is homogeneous and completely saturated.
- Soil grains and the water are incompressible.
- Compression and flow are one-dimensional (vertical); lateral strain is zero.
- Darcy's law is valid; k is constant during consolidation.
- m_v (and hence c_v) is constant; the relation between e and σ' is linear (for small stress increments).
- The strains are small; the load is applied instantaneously and stays constant; time lag is due entirely to the low permeability of soil (no secondary compression).

Differential equation

For an element of thickness dz, the rate of volume decrease = net outflow of water: $\partial(e)/\partial t$ ... leads to

$$\partial u / \partial t = c_v \, \partial^2 u / \partial z^2 \qquad c_v = k / (m_v \gamma_w) = k (1 + e_0) / (a_v \gamma_w)$$

- c_v = coefficient of consolidation (cm^2/s or $m^2/year$): indicates how fast the pore pressure dissipates. Large for sand and silt, small for clay (10^{-3} to 10^{-4} cm^2/s).
- Solution by Fourier series with $u = u_0$ at $t = 0$, and $u = 0$ at the drainage boundaries:

$$u = \Sigma \, (2 u_0 / M) \sin (M z / H_{dr}) \, e^{-M^2 T_v} \qquad M = (\pi/2)(2m + 1), \; m = 0, 1, 2, \ldots$$

- Time factor $T_v = c_v t / H_{dr}^2$ (dimensionless). H_{dr} = longest drainage path = H/2 for double drainage (sand above and below) and H for single drainage (one drainage face).

Degree of consolidation

- $U_z = (u_0 - u_z) / u_0 = 1 - u_z / u_0$ at a depth z. Average degree of consolidation U (whole layer):

$$U = S_t / S_c = 1 - \Sigma\, (2/M^2)\, e^{-M\hat{}\{2\}\, T_v\}}$$

- U depends only on T_v (a single curve for a given drainage and initial pressure distribution).
- Approximations: for $U < 60$ %, $T_v = (\pi/4)\, U^2$; for $U > 60$ %, $T_v = 1.781 - 0.933 \log_{10} (100 - U\%)$.

U (%)	10	20	30	40	50	60	70	80	90	100
T_v	0.008	0.031	0.071	0.126	0.197	0.287	0.403	0.567	0.848	infinity

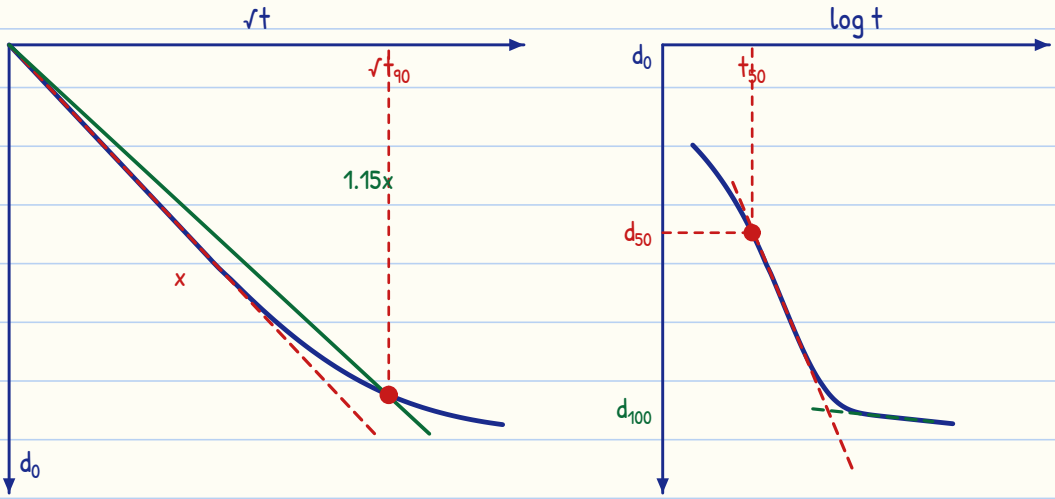
- Settlement at time t: $S_t = U \times S_c$. Time: $t = T_v\, H_{dr}^2 / c_v$. Time for the consolidation varies with the square of the drainage path; a layer with double drainage consolidates 4 times faster than with single drainage.

Example 5

Clay layer 4 m thick, double drainage, $c_v = 4 \times 10^{-3}$ cm^2/s, $S_c = 139$ mm.

- $H_{dr} = 2$ m $= 200$ cm. For $U = 90$ %, $T_v = 0.848$: $t_{90} = 0.848 \times 200^2 / (4 \times 10^{-3}) = 8.48 \times 10^6$ s = 98 days.
- For $U = 50$ %, $T_v = 0.197$: $t = 0.197 \times 40000 / 0.004 = 1.97 \times 10^6$ s = 22.8 days; $S_t = 0.5 \times 139 =$ 69.5 mm.
- If single drainage, $H_{dr} = 4$ m and $t_{90} = 4 \times 98 =$ 392 days.

8. Coefficient of Consolidation c_v - Curve Fitting Metho



Square root of time method *Log time method*

The c_v is found from the laboratory time - compression readings (dial reading vs time) of a load increment by two fitting methods, which match the theoretical U - T_v curve with the experimental curve.

(a) Square root of time method (Taylor, 1942)

- Plot the dial reading (compression) versus the square root of time ($min^{0.5}$). The initial part is a straight line ($U < 60$ %, $T_v = (\pi/4)U^2$, so the compression is proportional to sqrt(t)).
- Extend the initial straight line back to $t = 0$ to find the corrected zero d_0.
- Draw a second line from d_0 with all abscissae 1.15 times those of the first line. It cuts the experimental curve at the point corresponding to 90 % consolidation, which gives sqrt(t_{90}).

$$c_v = 0.848\ H_{dr}^2 / t_{90}$$

- (Since the theoretical ratio of abscissa at 90 % to the line is 1.15.) $d_{100} = d_0 + (10/9)(d_{90} - d_0)$.

(b) Logarithm of time method (Casagrande)

- Plot the dial reading versus log t. The curve is S-shaped (reverse) with a straight line in the middle part.
- Corrected zero d_0: choose two times t_1 and $t_2 = 4\ t_1$ (in the initial parabolic part); the vertical difference x between the readings gives the zero: $d_0 = d_1 - x$ (the reading at t_1 minus x).

- d_{100} = ordinate at the intersection of the tangent to the middle straight portion and the tangent to the final (secondary) portion.
- $d_{50} = (d_0 + d_{100})/2$; the time at d_{50} is t_{50}.

$$c_v = 0.197\ H_{dr}^2 / t_{50}$$

Calculations with the results

- H_{dr} is the average drainage path of the specimen for the load increment = half the average thickness (double drainage): $H_{dr} = (H_i + H_f)/4$.
- Then $k = c_v\ m_v\ \gamma_w$ - gives the coefficient of permeability of the clay.

Example 6 (root time)

Specimen 20 mm thick, double drained, H_{dr} = 1.0 cm. $\sqrt{t_{90}}$ = 3.4 $\sqrt{}$min, so t_{90} = 11.56 min = 694 s.

$c_v = 0.848 \times (1.0)^2 / 694 = 1.22 \times 10^{-3}$ cm^2/s.

Example 7 (log time)

t_{50} = 3.0 min = 180 s, H_{dr} = 1.0 cm: $c_v = 0.197 \times 1 / 180 = 1.09 \times 10^{-3}$ cm^2/s.

Comparison

Square root of time (Taylor)	Log of time (Casagrande)
uses t_{90}, T_v = 0.848	uses t_{50}, T_v = 0.197
gives a good initial straight line	needs the secondary compression to be clear
better for soils with little secondary compression	better for soils showing a distinct secondary portion
$c_v = 0.848\ H^2/t_{90}$	$c_v = 0.197\ H^2/t_{50}$

9. Secondary Consolidation

After the primary consolidation (u = 0), the soil continues to compress slowly at constant effective stress due to plastic adjustment (creep) of the soil grains and the adsorbed water layers.

- On the log t plot it appears as a straight line of small slope after d_{100}.

$$C_\alpha = \Delta e / \Delta \log_{10} t \qquad S_s = [\ C_\alpha\ H / (1 + e_p)\ ] \log_{10} (t_2 / t_1)$$

- e_p = void ratio at the end of the primary consolidation; t_1 = time at the end of primary; t_2 = time at which S_s is required.
- C_α / C_c = 0.03 - 0.06 for most clays, up to 0.1 for organic soils and peat. S_s is large for organic soils, peat, and highly plastic soft clays; small for over-consolidated clays.

- Secondary compression is not related to the dissipation of pore pressure (Terzaghi's theory neglects it).

10. Quick Revision

Item	Formula
Void ratio	$e = e_0 - \Delta H (1 + e_0) / H_0$
Coefficient of compressibility	$a_v = - \Delta e / \Delta \sigma'$
Volume compressibility	$m_v = a_v / (1 + e_0)$
Compression index	$C_c = - \Delta e / \Delta \log \sigma' = 0.009 (w_L - 10)$
Settlement (NC)	$S_c = [C_c H / (1 + e_0)] \log[(\sigma'_0 + \Delta\sigma)/\sigma'_0]$
OCR	$OCR = \sigma'_c / \sigma'_0$
Coefficient of consolidation	$c_v = k / (m_v \gamma_w)$
Time factor	$T_v = c_v t / H_{dr}^2$
Square root time	$c_v = 0.848 H_{dr}^2 / t_{90}$
Log time	$c_v = 0.197 H_{dr}^2 / t_{50}$
Secondary settlement	$S_s = [C_\alpha H / (1 + e_p)] \log (t_2/t_1)$

Important questions (exam)

- Distinguish between compaction and consolidation. Explain the spring analogy for primary consolidation.
- Explain immediate, primary and secondary settlements.
- Explain e - p and e - log p curves. Define a_v, m_v, C_c, C_s.
- What is stress history? Explain NC, OC and UC clays. How do you find the preconsolidation pressure?
- State the assumptions of Terzaghi's theory and derive the equation $\partial u/\partial t = c_v \partial^2 u/\partial z^2$.
- Describe the square root of time and the log of time fitting methods for c_v.

GEOTECHNICAL ENGINEERING

UNIT - V : SHEAR STRENGTH OF SOILS

Course Code: 23A01503T

Syllabus: SHEAR STRENGTH OF SOILS: Importance of shear strength - Mohr's-Coulomb failure theories - Types of laboratory tests for strength parameters - Strength tests based on drainage conditions - Critical void ratio - Liquefaction.

1. Importance of Shear Strength

Shear strength of a soil is the maximum resistance that the soil offers to shear stresses (sliding of one part of the soil mass over the other) before it fails. Soil fails in shear, never in compression or tension - so the strength of soil means its shear strength.

- Bearing capacity of foundations - the soil below the footing fails by shearing.
- Stability of slopes - embankments, earth dams, cuts, natural hill slopes.
- Earth pressure on retaining walls, sheet piles, bridge abutments, braced cuts.
- Stability of pavements, subgrade and the design of tunnels, pipelines and underground structures.
- Safety of excavations, the stability of the bases of the pile foundations and the skin friction of piles.

Sources of shear strength in a soil

- Cohesion (c) - the attraction between the particles of fine-grained soils (clay), due to electro-chemical bonds and adsorbed water, and cementing agents. It is independent of the normal stress.
- Internal friction (φ) - the frictional resistance between the grains: (i) sliding friction at contacts, (ii) interlocking of particles (must dilate to move over each other), (iii) rolling friction. It is proportional to the normal stress on the plane.

Clean sand and gravel: $c = 0$ (cohesionless, "frictional" soil). Saturated clay in undrained condition: $\varphi = 0$ (purely cohesive soil). Most natural soils (silty clay, clayey sand) are c - φ soils.

2. Stresses on a Plane - Mohr's Circle

In a stressed soil element, three mutually perpendicular planes have no shear stress on them - these are the principal planes, and the normal stresses on them are principal stresses: major σ_1, intermediate σ_2, minor σ_3. For two-dimensional problems σ_2 is ignored.

Analytical expressions

On a plane inclined at θ to the major principal plane (angle θ is measured from the plane carrying σ_1):

$$\sigma_n = (\sigma_1 + \sigma_3)/2 + (\sigma_1 - \sigma_3)/2 \cos 2\theta$$

$$\tau = (\sigma_1 - \sigma_3)/2 \sin 2\theta$$

- Maximum shear stress $\tau_{max} = (\sigma_1 - \sigma_3)/2$ at $\theta = 45^\circ$ (radius of the circle).

Mohr's circle (graphical)

- Take σ on the x - axis and τ on the y - axis (compression positive in soil mechanics).
- Mark σ_3 and σ_1 on the axis; the circle having the diameter $(\sigma_1 - \sigma_3)$ is the Mohr's circle. Centre at $(\sigma_1 + \sigma_3)/2$, radius $(\sigma_1 - \sigma_3)/2$.
- Every point on the circle gives the (σ, τ) on one plane. A plane at θ is represented by the point at an angle 2θ from the σ_1 end, measured at the centre.
- The pole (origin of planes) method: draw through the point of σ_3 a line parallel to the plane of σ_3, the intersection with the circle is the pole; the lines from the pole to any point of the circle give the orientation of the plane.

Example 1

Major and minor principal stresses at a point are 400 kN/m^2 and 100 kN/m^2. Find the normal and shear stress on a plane inclined at 30° to the major principal plane.

- $\sigma_n = (400+100)/2 + (400-100)/2 \cos 60^\circ = 250 + 150(0.5) = 325$ kN/m^2
- $\tau = (400-100)/2 \sin 60^\circ = 150 \times 0.866 = 129.9$ kN/m^2; $\tau_{max} = 150$ kN/m^2 (at 45°).

3. Mohr - Coulomb Failure Theory

Coulomb (1776) proposed that the shear strength of soil on a plane is a linear function of the normal stress on it:

$$\tau_f = c + \sigma \tan \varphi$$

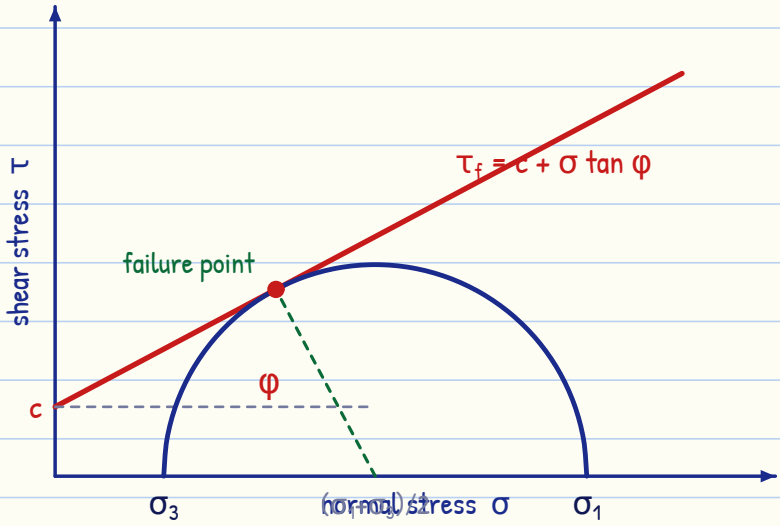
where τ_f = shear strength, c = cohesion intercept, φ = angle of shearing resistance (angle

of internal friction), σ = normal stress on the failure plane.

- Terzaghi's effective stress principle - the shear strength is controlled by the effective normal stress, hence:

$$\tau_f = c' + (\sigma - u) \tan \varphi' = c' + \sigma' \tan \varphi'$$

- Mohr (1900) postulated that a material fails when the shear stress on the failure plane reaches a function of the normal stress; the failure envelope is a curve. For the stress range met in practice the envelope is approximated by a straight line (Coulomb), hence the Mohr - Coulomb failure criterion.



Failure plane makes
$\theta = 45° + \varphi/2$
with the major principal plane;
the angle $2\theta = 90° + \varphi$
is measured at the centre.

Circle touches the envelope at failure; if it stays below the envelope the soil is safe.

Conclusions from the Mohr - Coulomb diagram

- The failure envelope is the tangent to all the Mohr's circles at failure. If the circle lies below the envelope, the soil is stable; if it touches, it is at failure; a circle above the envelope is impossible (the soil would already have failed).
- Failure takes place on the plane at $\theta_f = 45^\circ + \varphi/2$ to the major principal plane (not on the plane of maximum shear stress, 45°, as the friction acts also).
- The normal and shear stresses on the failure plane: $\sigma_f = (\sigma_1+\sigma_3)/2 + (\sigma_1-\sigma_3)/2 \cos 2\theta_f = (\sigma_1+\sigma_3)/2 - (\sigma_1-\sigma_3)/2 \sin \varphi$, $\tau_f = (\sigma_1-\sigma_3)/2 \cos \varphi$.

Relation between the principal stresses at failure

From the right-angled triangle of the Mohr's circle (centre, tangent point and the point where the envelope meets the σ axis at -c cot φ):

$$\sin \varphi = (\sigma_1 - \sigma_3) / (\sigma_1 + \sigma_3 + 2c \cot \varphi)$$

$$\sigma_1 = \sigma_3 \tan^2(45^\circ + \varphi/2) + 2c \tan(45^\circ + \varphi/2)$$

$$\sigma_3 = \sigma_1 \tan^2(45^\circ - \varphi/2) - 2c \tan(45^\circ - \varphi/2)$$

- $N_\varphi = \tan^2(45^\circ + \varphi/2)$ is called the flow value. For sands ($c = 0$): $\sigma_1/\sigma_3 = N_\varphi$, $\sin\varphi = (\sigma_1 - \sigma_3)/(\sigma_1 + \sigma_3)$.
- For purely cohesive soils ($\varphi = 0$): $\sigma_1 = \sigma_3 + 2c$, i.e. $c = (\sigma_1 - \sigma_3)/2 = \tau_{max}$ and the failure plane is at 45°.

Example 2

(a) A dry sand with $\varphi = 30^\circ$ is tested in triaxial with $\sigma_3 = 100$ kN/m^2. Find σ_1 at failure.

- $\sigma_1 = \sigma_3 \tan^2(60^\circ) = 100 \times 3 = 300$ kN/m^2; the plane of failure is at $45^\circ + 15^\circ = 60^\circ$ with the major principal plane.

(b) A clay with $c = 20$ kN/m^2, $\varphi = 20^\circ$, $\sigma_3 = 100$ kN/m^2:

- $\tan(55^\circ) = 1.428$, $\tan^2(55^\circ) = 2.039$; $\sigma_1 = 100(2.039) + 2(20)(1.428) = 203.9 + 57.1 = 261$ kN/m^2.

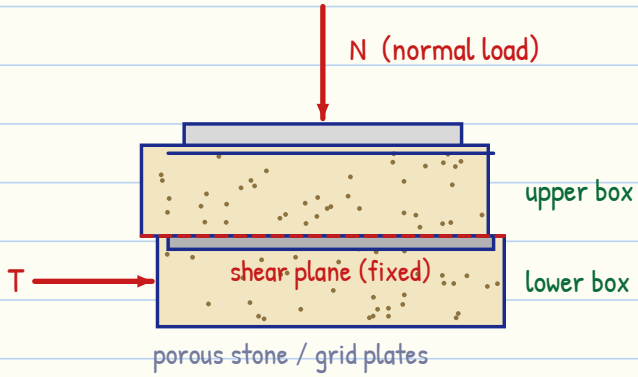
4. Types of Laboratory Tests for Strength Parameters

The shear strength parameters c and φ are determined from:

Test	Type of stress	Used for
Direct shear test (shear box)	failure plane is forced to be horizontal	sands, gravels, quick test on all soils
Triaxial compression test	all round pressure and axial load	all soils, any drainage condition (most versatile)
Unconfined compression test (UCC)	$\sigma_3 = 0$, special case of triaxial	saturated cohesive soils (clay), only c_u
Vane shear test	torque on the vanes	soft clays and silts (lab and field)

- Other tests: simple shear, ring shear, plane strain and torsion tests (research), and in-situ tests such as SPT, CPT and field vane.
- Selection of the test is made on the basis of the soil type, the available time and the field drainage conditions.

5. Direct Shear Test (IS 2720 Part 13)



Box 60 x 60 mm (or 100 x 100 mm)
Rate of shearing 1.25 mm/min
(sand 0.2 - 1 mm/min)

$\tau = T / A$, $\sigma = N / A$
Plot τ_f against σ at three values of N -> envelope:
slope = φ, intercept = c

Procedure

- The soil sample (sand compacted to the required density, or clay at the required water content) is placed in the shear box between two porous stones with grid plates (for drained test). The box is split horizontally into two halves, so the shear plane is predetermined.
- A vertical normal load N is applied through the loading pad; then the lower box is moved at a constant rate (strain controlled), the upper box being held by the proving ring, which measures the shear force T. Horizontal and vertical dial gauges record the displacement (and the volume change).
- The test is carried out on at least three identical specimens under different normal stresses (say 50, 100, 150 kPa), and the maximum shear stress (peak), $\tau_f = T_{max}/A$ is found for each. A (area) is corrected for the reduction in the shear plane as shearing proceeds (optional).
- Plot τ_f against σ: a straight line; slope = φ, intercept on the τ axis = c.

Types of direct shear test

- Undrained (quick) test - shearing begins immediately after applying the load (about 1 - 2 min), no time for drainage; Consolidated undrained - shear after consolidation under N, with the drainage closed; Drained (slow) test - the soil is allowed to consolidate under N, then sheared slowly (so that the pore pressure is nil): φ_d, c_d.
- For cohesionless soils the drained test is quick since drainage is free. For clay, the shearing time is $t_f = 12.7\ t_{100}$ (t_{100} from the consolidation) - several days.

Advantages and disadvantages

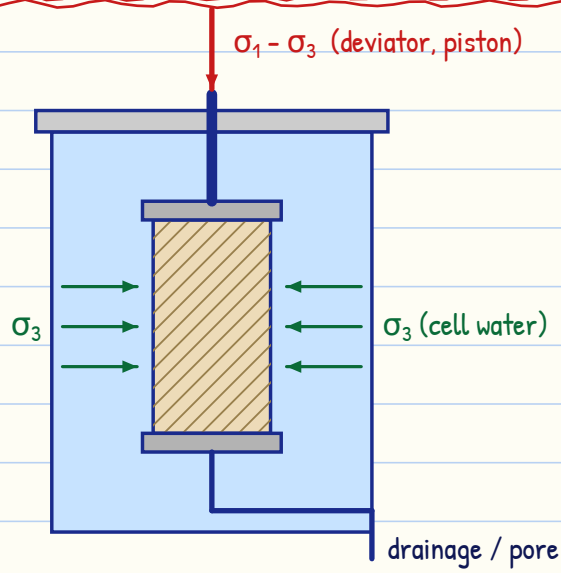
Advantages	Disadvantages
simple, quick and cheap equipment	the failure plane is forced (not the weakest plane)
easy to prepare the specimen, good for sands	non-uniform stress distribution, stress concentration at the edges
thin specimen: quick drainage and consolidation	no control on drainage; pore pressure not measured (only the total stress)
the residual strength can be found by reversal of the box	area of the shear plane changes during the test

Example 3

Direct shear on a dry sand (60 x 60 mm box) gives shear forces at failure of 208, 417 and 625 N for the normal loads of 360, 720 and 1080 N.

- Area = 36 cm^2 = 3.6 x 10^{-3} m^2. σ = 100, 200, 300 kN/m^2; τ_f = 57.8, 115.8, 173.6 kN/m^2.
- tan φ = 173.6/300 = 0.579 so φ = 30°, and the intercept c = 0 (dry sand).

6. Triaxial Compression Test (IS 2720 Part 11, 12)



Cylindrical specimen, L / D = 2
(38 mm x 76 mm, or 50 x 100 mm)
Rubber membrane, porous stones,
cell fluid gives all round σ_3.

Axial load raises σ_1 until failure
$\sigma_1 = \sigma_3$ + (deviator stress)

The most reliable and versatile test: the stress conditions are controlled, drainage can be controlled and the pore pressure measured. The specimen is surrounded by a fluid of pressure σ_3 (cell pressure), then the vertical stress is increased by the axial load through the piston up to failure: $\sigma_1 = \sigma_3 + \sigma_d$.

Procedure

- The specimen (38 mm dia x 76 mm length, or 50 x 100 mm; L/D = 2) is covered with a rubber membrane, put between the porous stones (filter paper strip along the side for quicker drainage), sealed with O-rings on the pedestal and the top cap and enclosed in

the cell filled with water.

- Cell pressure σ_3 is applied (and the specimen is saturated by back pressure if required). The axial strain rate is kept at 0.5 - 1.25 % per minute (undrained) or lower for drained tests (so that excess pore pressure dissipates).
- The load (proving ring) and the deformation are noted until the peak deviator stress or 15 - 20 % axial strain. The cross-section area is corrected: $A = A_0 / (1 - \varepsilon)$.

$$(\sigma_1 - \sigma_3)_f = P / A, \quad A = A_0 / (1 - \varepsilon)$$

- The Mohr's circles for 3 values of σ_3 are drawn and the common tangent gives c and φ.

Advantages over the direct shear test

- Drainage conditions and the pore water pressure are controlled / measured; the specimen fails along the weakest plane; the stress distribution is uniform; the specimens can be saturated; the effective stress parameters can be obtained; the complete stress path is available.
- Disadvantages: costly and elaborate equipment, skilled operator, long testing time (CD test on clay may take days), and $\sigma_2 = \sigma_3$ (axi-symmetric) only.

Example 4

CD triaxial tests on a soil give $\sigma_1 = 473$ kN/m^2 when $\sigma_3 = 100$ kN/m^2 and $\sigma_1 = 773$ kN/m^2 when $\sigma_3 = 200$ kN/m^2. Find c and φ.

- Subtract: $300 = 100\ N_\varphi$ so $N_\varphi = 3$, $\tan(45^\circ + \varphi/2) = 1.732 \rightarrow 45^\circ + \varphi/2 = 60^\circ \rightarrow \varphi = 30^\circ$.
- $\sigma_1 = \sigma_3 N_\varphi + 2c\sqrt{N_\varphi}$: $473 = 300 + 2c(1.732) \rightarrow c = 173/3.464 = 50$ kN/m^2.

7. Unconfined Compression Test (IS 2720 Part 10)

A special case of the UU triaxial test with $\sigma_3 = 0$. The cylindrical specimen (38 mm x 76 mm) of saturated cohesive soil is loaded axially in a compression machine (strain 1.25 %/min) until failure. The unconfined compressive strength q_u is the maximum axial stress (or the stress at 20 % strain).

- Failure circle passes through the origin: $\sigma_3 = 0$, $\sigma_1 = q_u$; as $\varphi_u = 0$ for the saturated clay, the envelope is horizontal.

$$c_u = q_u / 2 \quad (= \text{undrained shear strength } s_u), \quad \text{failure plane at } 45^\circ$$

- The test is used only for saturated clays (it can not be used for sands, dry soils, fissured clays or partially saturated soils, as they cannot stand without confinement).
- Consistency by q_u: very soft < 25 kPa, soft 25 - 50, medium 50 - 100, stiff 100 - 200, very stiff 200 - 400, hard > 400 kPa.

Sensitivity

S_t = (q_u undisturbed) / (q_u remoulded) at the same water content

- S_t = 1 - 2 insensitive; 2 - 4 normal; 4 - 8 sensitive; 8 - 16 extra sensitive; > 16 quick clay. Loss of strength on remoulding is due to the destruction of the structure; thixotropy is the regain of the strength with time at constant water content (e.g. piles driven in clay).

Example 5

An undisturbed clay specimen has q_u = 120 kN/m^2; remoulded at the same water content q_u = 30 kN/m^2.

- c_u = 120/2 = 60 kN/m^2; S_t = 120/30 = 4 (sensitive clay).

8. Vane Shear Test (IS 2720 Part 30)

For very soft to medium clays and sensitive clays which are difficult to sample; the in-situ test is done in the field by pushing a four-bladed vane (d = 75 mm, h = 150 mm; h = 2d) to the required depth and rotating it at 0.1°/s (6° per min) by a torque head.

- The soil is sheared along a cylindrical surface of the diameter d and height h, and the end circular areas. Torque at failure T = (resistance of the cylindrical surface) + (resistance of two ends):

$$T = \pi d^2 h s/2 + \pi d^3 s/6 \quad \rightarrow \quad s = T / [\, \pi (d^2 h/2 + d^3/6) \,]$$

- For the vane with only the top end in shear: $d^3/12$ instead of $d^3/6$. The vane shear gives the undrained strength (c_u) of the clay ($s = c_u$); the residual (remoulded) strength is found by rotating the vane rapidly several times.
- Advantages: quick, cheap, in-situ, the least disturbance, gives the sensitivity.
  Limitations: not suitable for stiff clays, sands and gravels; the strength is anisotropic; the rate effect and the vane insertion disturbance.

Example 6

Vane: d = 75 mm, h = 150 mm; the torque at failure T = 40 N-m.

- $d^2h/2 = (0.075)^2(0.15)/2 = 4.219 \times 10^{-4}$ m^3; $d^3/6 = 7.03 \times 10^{-5}$ m^3; sum = 4.922×10^{-4}.
- $s = 40 / (\pi \times 4.922 \times 10^{-4}) = 40/1.546 \times 10^{-3} = 25.9$ kN/m^2. So c_u = 25.9 kPa (soft clay).

9. Strength Tests Based on Drainage Conditions

The strength of a clay is not a constant: it depends on the drainage during loading. Three tests are carried out in the triaxial equipment, depending on the field conditions:

Test	Stage 1 (σ_3 applied)	Stage 2 (shear)	Result
UU Unconsolidated Undrained (Quick)	drainage closed, no consolidation	drainage closed	total stress: c_u, φ_u
CU Consolidated Undrained	drainage open, consolidates	drainage closed (u measured)	c_{cu}, φ_{cu}; c', φ' with u
CD Consolidated Drained (Slow)	drainage open, consolidates	drainage open, slow shear (u = 0)	c_d, $\varphi_d = c'$, φ'

(a) UU test

- Saturated clay: an increase of the cell pressure is taken up by the pore water entirely (B = 1), so the effective stress does not change, and the strength is independent of σ_3: all the circles have the same diameter and the envelope is horizontal:

$$\varphi_u = 0, \quad c_u = (\sigma_1 - \sigma_3)_f / 2$$

- Used for: end of construction stability of an embankment on soft clay, quick loading of a footing on clay, where the clay has no time to drain (short term - total stress analysis).
- For the partially saturated soil $\varphi_u > 0$ (the air compresses).

(b) CU test

- Specimen consolidated under σ_3 (drainage open), then sheared undrained. The pore pressure u is measured, hence the effective stress circles can also be drawn: $\sigma_1' = \sigma_1 - u_f$, $\sigma_3' = \sigma_3 - u_f$. The effective envelope gives c', φ' ($\varphi' > \varphi_{cu}$).
- Used for: rapid draw down of the earth dams, loading after the clay has consolidated (the next stages of construction), the strength gain with consolidation.

(c) CD test

- The soil is consolidated and then sheared so slowly that no excess pore pressure develops: total stress = effective stress. It gives the effective stress parameters c_d (c' ~ 0 for NC clay) and φ_d.

- Used for: long-term stability (the steady state after many years - cuts, natural slopes, retaining wall backfill), the sands (always drained), and the design of foundations on free-draining soils. The test takes a long time in clay.

Pore pressure parameters (Skempton)

$$\Delta u = B\,[\,\Delta\sigma_3 + A\,(\Delta\sigma_1 - \Delta\sigma_3)\,]$$

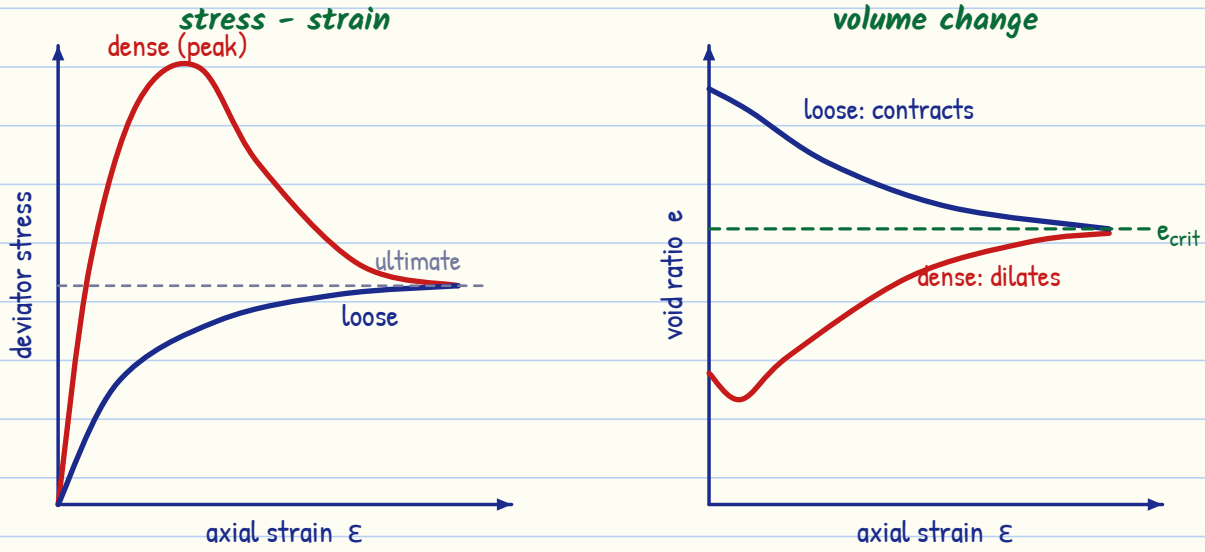
- B = $\Delta u / \Delta\sigma_3$ for the isotropic loading: B = 1 for the saturated soil, B = 0 for dry soil (the check of the saturation, B ≥ 0.95).
- A = the pore pressure parameter at the shearing: A_f = 0.5 - 1.0 normally consolidated clays, 0.25 - 0.5 lightly OC clay, 0 to -0.5 heavily OC clay (negative u due to dilation), 1 - 3 for sensitive clay. Dense sand A is negative.

Example 7

A saturated clay in the CU test: σ_3 = 150 kN/m^2, the deviator stress at failure = 130 kN/m^2, pore pressure at failure u_f = 70 kN/m^2. Find the effective principal stresses and A_f.

- σ_1 = 150 + 130 = 280; σ_1' = 280 - 70 = 210 kN/m^2; σ_3' = 150 - 70 = 80 kN/m^2.
- With B = 1: 70 = 1[0 + A(130)] -> A_f = 0.54; sin φ' = (210-80)/(210+80) = 0.448 so φ' = 26.6°.

10. Shear Behaviour of Sands - Critical Void Ratio



Dense and loose sand

- Dense sand (D_r high, low void ratio): the stress rises to a sharp peak at small strain, then falls to an ultimate value. The volume first decreases slightly, then increases (dilatancy) - the grains have to ride over one another (interlocking). Pore pressure in the undrained shear is negative.
- Loose sand: the stress rises slowly to the ultimate value without a peak and the volume decreases (contraction) all through. In undrained shear the pore pressure is positive.
- Both the dense and the loose sand, at large strain, reach the same ultimate (critical) strength and the same void ratio. φ_{peak} (dense) > φ_{ult} (loose): $\varphi_{peak} = \varphi_{cv}$ + dilation angle ψ.

Critical void ratio (Casagrande, 1936)

Critical void ratio e_c is the void ratio at which the sand undergoes no volume change during shearing, at a given confining pressure: a sand with $e > e_c$ (looser) contracts while at $e < e_c$ (denser) dilates.

- e_c is not a constant - it depends on the confining pressure: it decreases when the confining pressure (σ_3) increases. It is determined by a series of drained triaxial / direct shear tests on the sand specimens at different initial void ratios under the same σ_3: plot the volume change ($\Delta V/V$) at failure against the initial e; e at $\Delta V = 0$ is e_c.
- Significance: a saturated loose sand ($e > e_c$) with a tendency to contract under shear (or vibration) produces an increase in pore pressure, decrease in effective stress, and ultimately a flow failure (liquefaction). A dense sand ($e < e_c$) dilates and gains the strength. The critical density (relative density about 40 - 50 %) divides the two.
- The idea of critical void ratio has now developed to the critical state of soil mechanics: at large strain the soil flows at constant stress, constant volume.

II. Liquefaction

Liquefaction is the phenomenon in which a saturated, loose, cohesionless soil loses its shear strength due to a sudden rise in the pore water pressure under the dynamic (cyclic) loading, such as an earthquake, and behaves like a viscous liquid.

Mechanism

- Shaking (or other fast loading) tends to compact the loose sand, but the water cannot drain out in the short time (undrained condition): the load is transferred to the pore water, the excess pore pressure builds up with each cycle.
- Effective stress $\sigma' = \sigma - u$. When the excess pore pressure becomes equal to the total overburden stress ($u = \sigma$), $\sigma' = 0$ and, since $\tau_f = \sigma' \tan \varphi = 0$, the soil has no shear strength and flows. Pore pressure ratio $r_u = u/\sigma'_0 = 1$.

Strength: $\tau_f = (\sigma - u) \tan \varphi \rightarrow 0$ as $u \rightarrow \sigma$

- The excess water moves upward and appears at the surface as sand boils (mud volcanoes); the ground sinks and large flows take place.

Conditions necessary (factors affecting)

Factor	Liquefaction is likely when
Soil type	uniform fine to medium sand, non-plastic silt (poorly graded, $C_u < 2 - 4$); not in clays or in gravels
Density	loose, $D_r < 50$ %, SPT N-value < 15 (corrected $(N_1)_{60} < 30$ for the sandy soils)
Saturation	below the ground water table, usually within the upper 15 - 20 m
Drainage	poor drainage, so the pore pressure cannot dissipate during loading
Earthquake	high intensity (magnitude > 5), long duration (many cycles), peak ground acceleration > 0.1 g
Confining pressure	low effective overburden stress (shallow depth)

Types

- Flow liquefaction: loose soil, static shear stress greater than the residual strength; large displacements (flow slides) - Lower San Fernando dam, 1971.
- Cyclic mobility: limited, progressive deformation of the medium-dense soils during cyclic loading; lateral spreading near water fronts.

Effects and examples

- Bearing failure (tilting and sinking of buildings: Niigata, Japan, 1964, apartment buildings tilted), floating up of buried tanks, manholes and pipelines, settlement, lateral spreading of the slopes and river banks, failure of the earth dams, and damage to the retaining walls and ports (Kobe 1995). In India: Bhuj (Gujarat) earthquake 2001, Bihar - Nepal 1934, Assam 1950.

Evaluation

- Simplified procedure (Seed and Idriss): factor of safety F.S. = CRR / CSR against liquefaction, where CSR (cyclic stress ratio) = $0.65 (a_{max}/g)(\sigma_v/\sigma_v') r_d$ is the demand, CRR (cyclic resistance ratio) is the capacity obtained from the SPT / CPT / lab cyclic

triaxial tests. F.S. < 1 (generally < 1.2 – 1.3) shows liquefaction.

Mitigation measures

- Densification: vibro flotation, dynamic compaction, compaction piles, blasting.
- Drainage: gravel (stone) columns, earthquake drains – quick dissipation of the pore pressure; lowering the water table.
- Grouting / chemical stabilisation, soil replacement; reinforcing: deep foundations (piles) through the liquefiable layer, mat foundations, structural measures.

12. Shear Strength of Clays – Other Points

- Normally consolidated (NC) clay: $c' = 0$, the envelope passes through the origin, $\varphi' = 20^\circ$ – 30° (reduces as the PI increases). Over consolidated (OC) clay: the envelope is curved up to the preconsolidation pressure, $c' > 0$ for small stresses, and the strength is much more than that of the NC clay at the same stress.
- Peak and residual strength: for stiff OC clays, the strength is reduced to the residual strength ($c_r \sim 0$, $\varphi_r < \varphi'$) after large displacement – used in the analysis of old landslides (ring shear test).
- Strength of the partially saturated soil: the suction (negative pore pressure) gives the extra apparent cohesion; vanishes on saturation (slope failure after rains).
- Factors affecting the shear strength: soil type, particle size, shape and gradation, density (void ratio), water content, drainage conditions, effective stress, stress history, rate of loading, structure and cementation, temperature.

Stress path

- The path of the point $(p, q) = ((\sigma_1+\sigma_3)/2, (\sigma_1-\sigma_3)/2)$ during the loading. The K_f line has the slope $\tan\alpha = \sin\varphi'$ and the intercept $a = c' \cos\varphi'$. The effective stress path of the NC clay in the CU test bends to the left, of the OC clay to the right.

13. Quick Revision

Item	Formula
Mohr - Coulomb	$\tau_f = c + \sigma \tan \varphi$; $\tau_f = c' + \sigma' \tan \varphi'$
Plane at θ (major plane)	$\sigma_n = (\sigma_1+\sigma_3)/2 + (\sigma_1-\sigma_3)/2 \cos 2\theta$, $\tau = (\sigma_1-\sigma_3)/2 \sin 2\theta$
Failure plane	$\theta_f = 45^\circ + \varphi/2$
Principal stresses at failure	$\sigma_1 = \sigma_3 \tan^2(45^\circ+\varphi/2) + 2c \tan(45^\circ+\varphi/2)$
For sand	$\sin \varphi = (\sigma_1-\sigma_3)/(\sigma_1+\sigma_3)$
UCC	$c_u = q_u/2$; $S_t = q_{u(und)} / q_{u(rem)}$
Vane shear	$s = T / [\pi (d^2h/2 + d^3/6)]$
Skempton	$\Delta u = B [\Delta\sigma_3 + A (\Delta\sigma_1 - \Delta\sigma_3)]$
Effective stress	$\sigma' = \sigma - u$; at liquefaction $u = \sigma$, $\sigma' = 0$
Liquefaction check	F.S. = CRR / CSR

Important questions (exam)

- Explain the importance of shear strength of soil. State and explain the Mohr - Coulomb theory of failure and show that $\theta = 45^\circ + \varphi/2$.
- Derive the relation between the major and minor principal stresses at failure in terms of c and φ.
- Describe the direct shear test: the procedure, merits and demerits. How are c and φ found?
- Explain the triaxial test. Differentiate between UU, CU and CD tests; state where each is used.
- What is the unconfined compression test? Define the sensitivity of clay. Explain the vane shear test.
- What is the critical void ratio? How does it affect the behaviour of sand during shear?
- What is liquefaction? Explain the factors that cause it and the measures to prevent it.
- Numerical: Mohr's circle problems, c and φ from the triaxial / direct shear data, c_u from q_u, vane shear.